Shato

Puku ye e sepelelanago le OctoStudio



Diprojeke tše 20. Diswantšho tša gago. Lentšu la gago. Khoutu ya gago.

Circuit Kids

O amogetšwe go Shato

Sedirišwa le sedirišwa se sengwe le se sengwe go Circuit Kids se swanetše go ba le puku ye e se sepelelanago. Puku ye e dira gore molao woo e be tsela ya baithuti ba mathomo: pele o tsebe tsela go app, ke moka o dire selo seo se nago le bohlokwa.

Ka ga kgatišo ye

- Babadi bao ba reretšwego: mengwaga ye 7-12, le molekane goba motho yo mogolo ge go nyakega. Bana ba bannyane ba ka tšea karolo ka go thala, go anela le go bapala ka thušo ya motho yo mogolo.
- Kanegelo: bana ba bo malome ba akanyeditšwego Kamogelo, Tumelo le Mpho, le Koko wa bona, ba sepela ka Limpopo. Shato ya thapo e sepela le bona go tloga kanegelong ya mathomo go ya go ya mafelelo. Kanegelo ya 1 e thoma kgorong ya Koko kgauswi le Polokwane. Mafelo a nnete le meetlo ye e phelago ga di bontšhwe e le dilo tša batho ba ba akanyeditšwego.
- Sebopego: karolo ya go sepela go app, ke moka diprojeke tše 20: kanegelo le go dira, ke moka khoutu le go leka. Projeke ya 20 e na le karolo ye nngwe ya go leka le go naganišiša.
- Sebopego sa puku: e dirilwe go balwa mogaleng goba tabletheng, goba go gatišwa go ithuta mmogo.
- Leina le laesense: Shato ke lentšu la Sepedi la noga ya python. Puku ye e na le laesense ya CC BY-NC-SA 4.0; bona tsebišo ya ditokelo tša kopi le laesense ka fase.

Puku ye ke eng, gomme ga se eng

Thulaganyo ya mathomo ya go ruta e ile ya hlahlobja ka kgaetšedi ya semmušo ya July 2026 ya OctoStudio 1.1.0 le ya godimo. Diprojeke tše tša mathomo ga se tša ka tša dirwa sedirišweng sa nnete goba le bana. Dipoelo tše di solofetšwego ke dikgopolo tša go leka, e sego dikganetšo tša gore teko e fetile.

Circuit Kids e mo go ngwadišweng. Dipotšišo: 072 123 1490 | info@circuitkids.co.za.

Ditokelo tša kopi le laesense

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O ka abelana le go fetola dilo tše ka bolokologi ge e se go dira tšhelete. Bolela Circuit Kids, kgokagantšha laesense gomme o swaye diphetogo dife goba dife. Abelana ditokišo tša gago ka laesense ye e swanago.

Diteng

Bala kanegelo pele. E tlogele e dule le wena. Ke moka dira selo se tee sa lefase leo se sepele, se bolele goba se arabe.

Hwetša dinao tša gago

Pele ga Projeke 1: lokišetša app, tseba diskirini, dira setaki sa gago sa mathomo, ithute go lokiša phošo, gomme o boloke kopi ye e šireletšegilego ya mošomo wa gago.

Diprojeke tše 20

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Motho yo mogolo o re thuša go thoma

Pele bana ba goroga

- Kenya OctoStudio go tšwa dikgokaganyong tša semmušo tša lebenkele la di-app go octostudio.org o diriša inthanete. Ntšhafatša pele ga sekolo; diaparo di nyaka kgatišo 1.1.0 goba e mpsha. Bula app gatee gomme o lekole puo ye e kgethilwego.
- Diriša mogala goba tablet ye e thekgwago: kgaetšedi ya bjale e ngwala Android 8+, iOS 15+, goba Chromebooks tše di nago le di-app tša Android. Dikhomphutha tše dingwe tša tlwaelo ga di thekgwe ke app ye. Lekola disedirišwa tša sekolo tša nnete pele o holofetša kharikhulamo.
- Ka morago ga go kenya, go hlama go šoma ntle le inthanete. Boloka sebaka sa diswantšho, modumo le difaele tša projeke. Go romela faele inthaneteng go ka nyaka data; go hlama ntle le inthanete ga se go abelana mahala inthaneteng.
- Dira babadi ka bobedi: yo mongwe ke moetapele yo a thintšago sekirini, yo mongwe ke moetapele yo a balago goba a supago. Fetolang projeke ye nngwe le ye nngwe. Ka fase ga mengwaga ye 7, diriša go thala, go anela le papadi ya boloko bja bo tee yeo e kaetšwego ke motho yo mogolo go e na le go solofela tatelano ka moka ka bobona.
- Go Gae > Dipeakanyo, bula Matšhwao a Mathomo a baithuti ba mathomo. Bontšha go thinta sekirini go thuša dipontšho. Leka VoiceOver goba TalkBack ya sedirišwa le medumo ya dibolokwa tša khoutu le moithuti yo a di nyakago. Leka gore o bone gore go kgonega, o se ke wa nagana gore peakanyo e tee e lekane bohle.

Kano ya rena ya polokego

Diriša diswantšho le medumo ya lebokoso la dipuku pele. Kgopela pele ga o tšea seswantšho sa motho goba o rekhota lentšu la gagwe. O se ke wa akaretša maina a dikolo, diaterese, dinomoro tša mogala goba dipoledišano tša poraefete diprojekeng. Motho yo mogolo o sekaseka se sengwe le se sengwe seo se tlogago sedirišweng. O se ke wa diriša app ye ge o tshela tsela. Swara disedirišwa ka go tia godimo ga selo se bonolo ge o sega goba o šikinya. Ga go na go šikinya ka maatla.

Molao wa poraefete wa app o re data e bolokwa selegaeng gomme ga e kgoboketšwe ke OctoStudio. Faele ye e romelwago go tirelo ye nngwe ke tšweletšo ye e ikemetšego. Boloka kopi pele o tloša app; go tloša go ka phimola data ya selegae ya app. Ga go nyakege go abelana phatlalatša go fetša puku ye.

Tšea OctoStudio mahala

Skena khoutu ya sedirišwa sa gago ka khamera, goba o ngwale aterese ya webosaete ka fase. App e dirilwe ke sehlopha sa MIT Lifelong Kindergarten (sehlopha sa Scratch) gomme dikgokagano di tšwa octostudio.org.

Android (Google Play):
https://play.google.com/store/apps/details?id=edu.mit.octostudio

iPhone le iPad (App Store):
https://apps.apple.com/app/octostudio/id6461573316

Tseba diskirini tše pedi

Gae: shelofo ya diprojeke tša gago

Gae e bontšha diprojeke tša gago. Hlama projeke ye mpsha e thoma ye mpsha. Nyakišiša e na le mehlala yeo o ka bapalago ka yona. Lekgasa le bula Dipeakanyo. Lenaneo la projeke le go nea Kopa, Rata, Abelana le Phimola. Leka Kopa projekeng ya go ikatisa, e sego Phimola mošomong wa gago o mobotse.

Mohlami: lebala la gago le lennyane

SETHALE: bona le go ama kanegelo ya gago

Play / Stop Oketša ka pela: sprite, motheo, modumo

LENANEO LA SPRITE: kgetha khoutu ya mang yeo o e fetolago

LEFELO LA KHOUTU: kgomaretša ditaelo tša gago

LENANEO LA DIBOLOKO: nyaka ditaelo

Mmapa wa mathomo wo o nolofaditšwego, e sego seswantšho sa sekirini. Go Chromebooks le ditabletheng tše kgolo dikarolo tše di swanago di ka beakanywa ka thoko ga nngwe.

Sprite ke sebopego goba selo seo o ka se ngwalelago khoutu. Motheo ke lefelo leo le lego ka morago ga sona. Sethale ke felo moo projeke e šomago gona. Boloko ke taelo e tee. Setaki, goba skripte, ke sehlopha seo se kgokagantšwego sa dibolokwa tša sprite e tee.

Thinta sprite lenaneong la disprite pele o e ngwalela khoutu. Sethale se bontšha kanegelo ka moka, eupša lefelo la khoutu ke la sprite ye e kgethilwego. Ge sebopego se se fošagetšego se sepela, lekola kgetho pele.

Dipuo tša Afrika Borwa go OctoStudio

Lenaneo la bjale la dipuo tša OctoStudio le akaretša Afrikaans, English, isiXhosa, isiZulu, Sepedi (Northern Sotho), Setswana (Tswana) le Tshivenḓa (Venda). Kgetha puo ya app go Gae > Dipeakanyo. Baithuti ba ka anela dikanegelo ka dipuo tšeo ba di dirišago gae; kgopela motho yo a di bolelago gore a lekole dikgumedišo le direkhoto.

Mothopo: https://octostudio.org/about | Lenaneo la dipuo le lekolotšwe ka 10 Diphalane 2026.

Dira taelo ya gago ya mathomo

- Hlama projeke ye mpsha go tšwa go Gae. Kgetha motheo go tšwa lebokosong la dipuku. Diriša Oketša sprite gomme o kgethe emoji goba Penta sebopego. Boloka sprite e tee fela go ikatiša.
- Kgetha sprite ya gago lenaneong la disprite. Nyaka mohuta wa "Go thoma neng?" lenaneong. Goga boloko bja Play-thoma lefelong la khoutu.
- Bula Motšamao. Goga boloko bja Tlola ka fase ga boloko bja go thoma go fihlela dikarolo di kgomarela. Thinta palo ya bjona gomme o leke 20.
- Gatiša Play ka fase ga sethale. Sprite ya gago e swanetše go tlola. Gatiša Stop. Bjale fetola 20 go ba 40 gomme o gatiše Play gape. Go fetogile eng?
- Goga boloko bja bobedi bja Tlola ka fase ga bja mathomo. Akanya gore dikgato tše pedi di tla bonagala bjang. Bapala gomme o bapetše. Ge go se na selo seo se diregago, kgokagantšha dibolokwa gape gomme o lekole sprite ye e kgethilwego.

BALA O YA FASE SETAKENG

Ge Play e thoma

Tlola 20

Tlola 40

Khoutu ke leano leo sedirišwa se ka le latelago. Boloko bja go thoma bo araba "Neng?" Dibolokwa tše di latelago di araba "Dira eng?" Go fetola palo e tee go fetola leano. Go leka ke go leka leano le go lebelela ka kelohloko.

Diresipi puku ye di diriša mantšu a tlhaloso a dibolokwa le dikgetho tša tšona tša lenaneo. Ga se khoutu ye e ngwadilwego ka tlhamalalo, diswantšho tša app goba kano ya gore puo e nngwe le e nngwe e diriša maina a a swanago. Diriša kgaetšedi ya semmušo go swantšha diswantšho tša dibolokwa. Methalo ye e tsentšwego e tsena molomong wa loop goba kgetho.

Lebokosa la gago la didirišwa tša dibolokwa

Go thoma neng?

Play, go thinta, go šikinya, go ama sprite, go fetola motheo. Ketsakgato e botša setaki gore se thome neng.

Motšamao

Sepela, tlola, retolla, ya felong la sethale, lebelela mothalong. Diriša go sepela go gonnyane pele ga go golo.

Mantšu le Medumo

Bubble ya polelo, mongwalo wa sethale, lentšu leo le bolelago, modumo wo o rekhotilwego goba wa lebokoso la dipuku. Letša modumo "go fihlela o fela" ge tiro ye e latelago e swanetše go leta.

Lefelo

Bogolo, diaparo, uta/bontšha, motheo, ka pele/ka morago. Diaparo ke dibopego tše di fapanego tša sprite e tee.

Mebala le Seetša

Phadima, diphetogo tša mmala, lebone. Puku ye e diriša seetša sa sekirini gore lebone le se nyakege.

Taolo

Boeletša, ka mehla, ge, leta, ema, romela le go amogela molaetša. Dibolokwa tše di beakanya tatelano.

Dibolokwa tše dingwe

Sega, dibolokwa tše di fetogago, dipalo, beam le dibolokwa tše di ikgethilego. Oketša go gakanega fela ge kanegelo ya gago e di nyaka.

Dibolokwa tše dingwe di na le dikgetho tša lebelo: nonne, magareng, ka pela le ka pela kudu. "Bea bogolo" e kgetha bogolo bjo bo feletšego; "fetola bogolo" e oketša goba e tloša go tšwa bogolong bja bjale. "Bea dintlha" e emiša palo; "fetola dintlha" e oketša go yona. Diphapano tše di bohlokwa ge o thoma gape.

Boloka mošomo wa gago. Lokiša diphošo tša gago.

Projeke ke ye ntši go feta vidio

- Feleletša phetogo ye nnyane, hlokomela Stop, ke moka o diriše Gae. Nyaka projeke ya gago lenaneong la Gae gomme o e bule gape. Lekola gore mošomo wa gago wa mafelelo o sa le gona sedirišweng se.
- Pele ga teko ye kgolo, diriša lenaneo la projeke > Kopa. Bula kopi gomme o e fetole. Boloka ya mathomo e šireletšegile.
- Go dira kopi ya go boloka, bula Abelana > Romela Faele ya Projeke. Kgetha kgetho ya go boloka faele sedirišweng ge e gona. Boloka kopi foldareng yeo motho yo mogolo a ka e hwetšago. Kgaetšedi e netefatša go boloka faele ya projeke sedirišweng; dikonope tša letlapa la go abelana di a fapana.
- Leka kopi: bula faele yeo ya .octostudio ye e bolokilwego ka OctoStudio gomme o lekole disprite le dibolokwa tša yona. Go amogela faele ya projeke go nyaka OctoStudio. Vidio goba GIF ke rekhoto fela, e sego khoutu ye e ka lokišwago.

Ge selo se sepela gampe

Ga go šome selo? Lekola boloko bja go thoma, kgokagano ye e kgomarelago le ketsakgato ye e nepagetšego. Sprite ye e fošagetšego? E kgethe lenaneong la disprite. Dibolokwa di nyametše ka ntle ga sekirini? Kgutšwafatša goba o beakanye lefelo la khoutu. Phetogo ye e fošagetšego? Leka Busetša morago, ke moka Dira gape ge go nyakega. Sprite ye e sa bonagalego? Lekola Uta le bogolo. Modumo wo o homotšego? Lekola kgetho ya modumo, bophara le ditumelelo. Ge ketsakgato e fetoga ka morago ga papadi e nngwe le e nngwe, busetša boleng bja yona bja mathomo setakeng sa Play.

Ema pele ga go busetša sprite mathomong. Gatiša Play gape go leka gape. O se ke wa phimola projeke go lokiša phošo ye nnyane. Botša molekane wa gago: "Ke be ke solofetše ___. Ke bone ___. Ke tla fetola ___."

Ikatiše go boloka kopi ntle le diswantšho tša poraefete goba medumo. Abelana kamoreng fela ge motho yo mogolo a dumela. Go boloka faele e šireletšegile ga go nyake go e bea inthaneteng.

PROJEKE 01

Thobela kgorong ya Koko



Kamogelo o be a eme kgorong a swere kgetsana ya gagwe ka matsogo a mabedi. Jarata e be e le kgolo go feta ka fao a e gopolago, le tšhušumetšo e be e le kgolo bjalo. O be a tlile Polokwane matsatsing a malofa a sekolo. Bana ba bomalome Tumelo le Mpho ba be ba boletše gore ba tla mo kgahlanetša. Ba be ba se gona.

Ke moka Tumelo a goroga ka sekhutlwaneng sa ntlo a swere se sengwe se se phadimago. Shato e nyenyane, e kobilwego ka thapo, e na le lebotlana la leihlo. "Thobela," a rialo, e sego go yena eupša go shato, a e dira gore e kokobele. Kamogelo a sega pele a ka itšhireletša.

Koko a tšwa a phumola diatla. A se a akgofa. A lebelela Kamogelo, a lebelela shato ya thapo, a lebelela kgoro. "Moeng ga se a swanela go ema moo," a rialo. Kamogelo a tsena ka gare.

Bosiu bjoo a thala shato ya thapo bukeng ya gagwe. Ge a e thinta sekirining, a e nea lentšu le tee. Lentšu leo a bego a sa kgone go le bolela kgorong. "Thobela."

Potšišo ya kanegelo: *Sebopego sa gago se ka bolela eng gore motho a ikwe a amogetšwe?*

Bokgoni: Go thinta, sprite, polelo

Dira

1. Hlama projeke e mpsha gomme o kgethe motheo o monyane wa lebokoso la dipuku. Penta sprite ya shato ya thapo e botse. Mmele o kobegilego o bonolo le sefahlego di lekane; ga go nyakege seswantšho sa phoofolo ya naga.

2. Kgetha sprite ya gago lenaneong la disprite. Goga boloko bja Thinta-thoma lefelong la khoutu. Kgetha kgetho yeo e bolelago go thinta sprite ye, e sego felo le felo.
3. Go Mantšu le Medumo, kgomaretša boloko bja bubble ya polelo. Ngwala 'Thobela, o amogetšwe' gomme o beye nako ya metsotswana e 2.

Khoutu (recipe)

```
SHATO YA GAGO YA THAPO YA GO DUMEDIŠA
When this sprite is tapped
Say in bubble: Thobela, welcome for 2 seconds
```

Go šoma bjang

Go thinta ke ketsakgato. E thoma setaki se fela ge sprite ye e kgethilwego e thintšwa. Bubble ke tiro. Ga go na seo se swanetšego go sepelela pele ga go thinta.

Leka

Gatiša Play ge go nyakega gore o tsene pono ya go sepela. Thinta sprite, ke moka karolo e se nago selo ya motheo. Ke sprite fela yeo e swanetšego go dumediša. Lekola kgetho ya go thinta ge ka bobedi di fetola.

Fetola

Ngwala dikgumedišo pampiring ka dipuo tšeo lapa la gago le di dirišago. Kgetha ye nngwe ya sprite ya gago. Kgopela motho yo a bolelago puo yeo gore a lekole mantšu; o se ke wa tshepa modumo wa mochini fela.

Hlalosa le go abelana

Bontšha molekane wa gago kamoo o kgethago sprite le go hwetša khoutu ya yona. Boloka kopi ya faele ya projeke ka morago ga go dumediša go go atlegilego la mathomo.

PROJEKE 02

Mogato o na le tatelano



Thekisi ya go tšwa Polokwane go ya Tzaneen e be e namela ka go kgotsofala. Ka morago ga Haenertsburg tsela e thoma go kgopama ka Magoebaskloof, gomme dihlare tša tswalana ka mahlakoreng ka bobedi. Mpho a bala dikhutlo ka go hlaboša lentšu gomme a lahlega ka bo lesome le bobedi.

Koko a be a pakile flaske ya tee gomme a ba botša molao. "Tseleng ye bjalo, re dira dilo ka tatelano. Pele khutlo, ke moka khutlo ye e latelago. Ga go yo a tlolago go ya fase." Mokgweranyi a dumela a sa retologe.

Kamogelo a leka go thala tsela go lengele la gagwe. A thala shato ya thapo pele gomme a bea tsela ka fase, shato ya fela moyeng. A thoma gape. Tsela pele, mathomo le bofelo. Ke moka shato, e sepela hlwahlwana ka hlwahlwana.

"Tatelano," Tumelo a rialo. Khutlong ye e latelago, senwelo sa tee seatleng sa Koko sa kgetha, gomme ga go yo a tšholotšego lerothi. Ka moka ba dumelelana gore mokgweranyi o swanetše go ba le sengwe.

Potšišo ya kanegelo: *Shato ya gago e swanetše go thoma le go felela kae leetong la yona?*

Bokgoni: Motšamao le tatelano

Dira

1. Oketša sprite ya shato goba o penta sebopego se bonolo se kobegilego se na le sefahlego. Diriša motheo wa lebokoso la dipuku. Ye ke polelo ya sekirini, e sego tiro ya go tshela tsela.

2. Kgetha shato. Kgomaretša boloko bja Eya-felong ka fase ga Play. Thinta pini ya yona ya mmapa gomme o bee shato kgauswi le lehlakore la ka ntle la sethaleng, kgole le lebopo.
3. Oketša Sepela go ya ka ntle go tliša 60 ka lebelo la magareng, ke moka bubble ya polelo 'Re fihlile!' ka metsotswana e 2. Tlogela lefelo le le lekanego la go sepela.

Khoutu (recipe)

```
SHATO TSELENG
When Play starts
Go to your chosen left-side start (instant)
Move right 60 (medium)
Say in bubble: We have arrived! for 2 seconds
```

Go šoma bjang

Bala o ya fase: pele busetša shato mathomong, ke moka sepela, ke moka bolela. Boloko bja Eya-felong bo kgetha felo. Sepela e fetola felo felo ka kgole. Go busetša go dira gore go sepela ka moka go thome felong le lengwe.

Leka

Bapala gabedi. Shato e swanetše go thoma felong le lengwe nako le nako. Fetola 60 go ba 30 gomme o bapetša. Ge e kopana le lebopo, šuthišetša mathomo ka gare goba o kgutšwafatše kgole.

Fetola

Thala leeto pampiring. Akanya kgole, o e leke, ke moka o e lokiše. Fetola motheo go ba felo kanegelong ya gago, o sa tšee seswantšho sa aterese.

Hlalosa le go abelana

Hlalosa gore ke boloko bofe bjo o ka bo šutišago ge shato e bolela pele e fihla.

PROJEKE 03

Mmete woo o retologago



Ka thoko, Mma Selina o be a loga mmete wa bjang lebatong la gagwe. A thoma ka sebopego se sennyane sa dikgopa tše mpsho le tše šweu. Ke moka a se retolla kotara, gomme a se dira gape. Gape gape. Ka pela mmete wo wa ba le sekwere sa dibopego tšeo di bego di swana le palesa, le naledi, le kgoro.

Mpho a lebelela a bea mogolo godimo ga mangole. "O tseba bjang gore o fedile?" Mma Selina a sega a sa lebelele godimo. "Ge o lebile gabotse bjalo ka ge o thomile."

Kamogelo a adima pensele. Letšhwao le lennyane le tee. Retolla letlakala kotara. Letšhwao. Retolla. Letšhwao. Retolla. Ka morago ga dikhutlo tše nne, letšhwao la gagwe la mathomo le be le mo letetše ka godimo.

Tumelo a nyaka go kobela shato ya gagwe ya thapo ka sebopego se se swanago. E be e sa retologe bjalo ka pampiri. E mo dira gore a nonne, gomme a hwetša gore go na le boithabišo. Koko a se bolele selo, eupša a bea tee kgauswi le bona gomme a dula.

Potšišo ya kanegelo: *Dikotara tše kae tša go retolla di dira go retolla ga botlalo?*

Bokgoni: Go thala le go boeletša

Dira

1. Oketša sprite > Penta. Thala sebopego se sennyane se se swanago le tše di lego mmeteng wa Mma Selina, sekwere goba daemane, ka sebopego goba borashe. Diriša Tlatša ge o nyaka mmala wo o tletšego. Feleletša seswantšho gore se bonale lenaneong la disprite.

2. Kgetha sebopego sa mokgwa. Ka fase ga Play oketša boloko bja Boeletša go tšwa go Taolo. Ngwala 4.
3. Bea Retolla ka go ya ka fa letsogong le letona 90 ka gare ga molomo wa Boeletša. Kgetha lebelo la magareng gore o bone go retolla ka go retolla.

Khoutu (recipe)

```
SEBOPEGO SA MOKGWA
When Play starts
Repeat 4 times:
    Turn clockwise 90 (medium)
```

Go šoma bjang

Boeletša e dira dibolokwa tšeo di lego ka gare ga yona makga a balwago. Dikhutlo tše nne tša dikhurii tše 90 di dira dikhurii tše 360. Boloko bja go retolla bo swanetše go ba ka molomong, e sego ka ntle ga Boeletša.

Leka

Akanya gore sebopego sa mokgwa se tla lebelela kae ka morago ga go retolla gatee. Bapala ka Boeletša 1, ke moka 2, ke moka 4. Mofuta wa dikhutlo tše nne o swanetše go fela o lebeletše bjalo ka ge o thomile.

Fetola

Penta mokgwa o mongwe sebopegong sa gago sa mokgwa. Leka Boeletša 6 ka dikhutlo tša dikhurii tše 60. Thala mokgwa pele o o diriša.

Hlalosa le go abelana

Dumelela molekane wa gago gore a bontšhe taelo yeo e boeletšwago. Loop e go phološa go goga boloko bjo bo swanago bja go retolla makga a mane.

PROJEKE 04

Mokgwa wa mphaka ga o akgofe



Kgauswi le Modjadjiskloof sethokgwa se lebane le thaba se be se bonala se farologane le sethokgwa se sengwe le se sengwe seo Kamogelo a se tsebago. Dihlare di be di na le dikutu tše kgolo tše di nago le boya le mogoga wa matlakala a thata a phadimago. "Dimokwa," Koko a rialo. "Dimokwa tša Modjadji. Polokelo ye e na le ye makgolo a dikete, gomme di gola gabohlale kudu." Mohlahli a dumela. Polokelo e rotšwe ka dikgošikgadi tša Balobedu tšeo di dulago lefelong le.

Kamogelo a nyaka go thala o tee ka pela. A gatelela kudu gomme kutu ya tlala letlakala. Tumelo a mmontšha tsela ye nngwe. A thala mokwa o monyane letlakaleng le tee, o mongwe o godimo hanyane o latelago, ke moka o mongwe.

Ge matlakala a bewa ka tatelano, sehlare se be se bonala se rotoga. Se senyane. Se segolo. Se segolo gape. Mpho a re sekirini se ka godiša mokwa ka moya o tee. "Wa nnete o tšea nako e telele kudu," Koko a rialo. "Ga a hlwe a ikwa."

Kamogelo a ngwala mantšu ao ka fase ga seswantšho sa gagwe, a nyenyane, gore a se ke a tšea letlakala ka moka.

Potšišo ya kanegelo: *Diphetogo tše dintši tše nnyane di ka anela kanegelo ya go gola bjang?*

Bokgoni: Bogolo le go ema

Dira

1. Penta sprite ya mokwa wa mofsa, goba o kgethe emoji ya semela bjalo ka sebopego se se nolofaditšwego gabotse. E kgethe.
2. Ka fase ga Play oketša Bea bogolo 30 lekgolong. Oketša Boeletša 4.
3. Ka gare ga Boeletša, bea Ema motsotswana o 1 wa latelwa ke Fetola bogolo ka 10. Diriša lebelo leo le bonalago, la magareng.

Khoutu (recipe)

```
MOKWA WA MOFSA
When Play starts
Set size to 30% (instant)
Repeat 4 times:
    Wait 1 second
    Change size by 10
```

Go šoma bjang

Bea bogolo e fa boleng bja mathomo. Fetola bogolo e oketša go bjona. Go thoma ka 30 gomme o oketša 10 makga a mane go dira 70 lekgolong. Ema e dumelela molebeledi go lemoga kgato ye nngwe le ye nngwe. Ye ke animeišene ya kanegelo, e sego tekanyo ya kgolo ya nnete ya mokwa.

Leka

Bapala gabedi gomme o lebelele bogolo. E swanetše go thoma e le nnyane makga ka bobedi. Tloša go busetša fela projekeng ye e kopilwego gomme o bone gore ke ka baka la eng go bapala makga a mantši go ka dira gore go gakanege.

Fetola

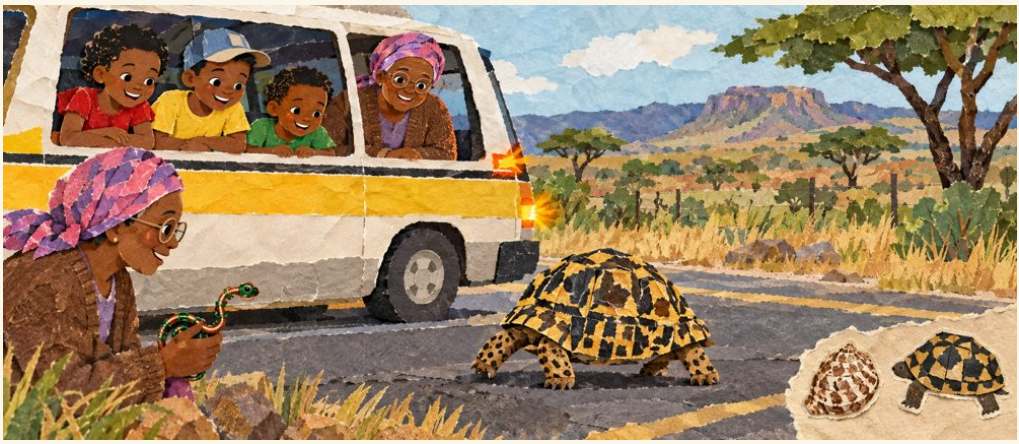
Dira gore mokwa wa mofsa o fokotšege gape ka go fetola palo go ba -10 loopong ya bobedi. Boloka bogolo bja mathomo e le bjo bogolo go lekana gore bo bonale.

Hlalosa le go abelana

Dira mothalo wa dipalo wa pampiri wa 30, 40, 50, 60, 70. Botša molekane wa gago gore sedirišwa se bala eng.

PROJEKE 05

Kgwara yeo e sa tsogego



Tseleng kgauswi le lepheng la Kruger, kgauswi le Phalaborwa, thekisi e ile ya ema ka baka la kgwara yeo e tshelago tara. E be e le kgwara ya nkwe, e na le matheba a matsho le a serolwana bjalo ka helemete ye kgale. Ga go yo a e amilego. Mokgweranyi a bea dilaete tša kotsi gomme a leta.

Ge Mpho a ikgoga kgauswi kudu, kgwara ya goga hlogo le maoto gomme ya nyamela. Ke legapi fela le sa šetšego. "A e sa le gona?" a hlabolla ka sephiri. Koko a re, "Leta." Ka morago ga nakwana hlogo ya tšwa gape, ka go kgotsofala, bjalo ka motho a hlaloša ka lepatleng.

Gae Kamogelo a thala diswantšho tše pedi. Legapi, le kgwara e na le maoto le hlogo. A bea ye nngwe godimo ga ye nngwe. Ge a thinta, legapi la šuthela ka thoko gomme kgwara ya bonala. Ge a thinta gape, ya ipata.

Go tšere dikgato tše mmalwa pele nako e ikwa e lokile. A rea projeke "E sa le gona."

Potšišo ya kanegelo: *Ke eng seo se swanetšego go bonala ge motheetši wa gago a hwetša leswao?*

Bokgoni: Uta, bontšha le disprite tše pedi

Dira

1. Oketša disprite tše pedi: kgwara ya nkwe yeo o e pentago le legapi la leswao. Di tlogele ka bobedi sethaleng. Bea legapi kgauswi le kgwara, e sego godimo ga yona, gore Bontšha e dire gore kgwara e bonale; legapi ke sepheo sa go thinta, e sego sekhurumelo.

2. Kgetha sprite ya kgwara. Aga setaki sa yona sa Play > Uta ka lebelo la ka pela. Oketša setaki se se arogilego sa Thinta-thoma sprite ye e tee, eupša o bee sepheo sa go thinta e le sprite ya legapi.

3. Ka fase ga ketsakgato yeo ya go thinta oketša Bontšha ka 100 lekgolong le bubble 'E sa le gona!' ka metsotswana e 2. Kgwara e na le ditiro le ge legapi le amogela go thinta.

Khoutu (recipe)

```
KGWARA: MATHOMO
When Play starts
Hide (instant)
```

```
KGWARA: GO BONTŠHA
When painted shell is tapped
Show at 100% (instant)
Say in bubble: Still here! for 2 seconds
```

Go šoma bjang

Sprite e tee e ka ba le ditaki tše dintši. Kgwara e iphihla ge Play e thoma gomme e bonala ka morago ga go thinta legapi. Sprite ye e phihlilwego ga e kgone go dirišana, ka gona o se ke wa nyaka gore kgwara ye e phihlilwego e amogele go thinta.

Leka

Bapala, leta gore kgwara e iphihle, ke moka o thinte legapi. Kgwara e swanetše go bonala. Ema o bapale gape: e swanetše go iphihla gape. Ge legapi le nyamela, o ngwadile khoutu go sprite ye fošagetšego.

Fetola

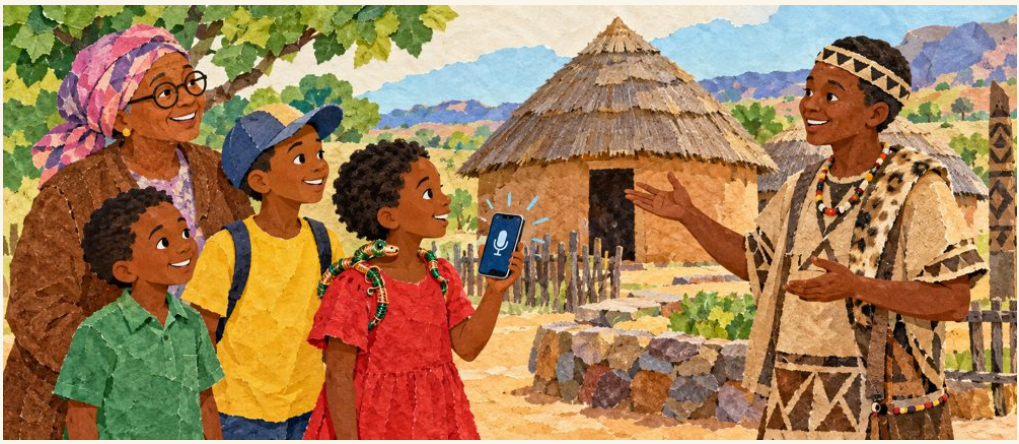
Thala leswao leo le nago le bohlokwa lapeng la gago. Dira gore leswao la bobedi le utolle selo se sengwe seo se pentilwego, tiro e tee ka nako.

Hlalosa le go abelana

Hlalosa gore ke ka baka la eng go uta leswao le tee fela leo le ka thintšwago go be go dira gore papadi e se kgonege.

PROJEKE 06

Botšiša mohlahli pele



Musiamong wa Bakone Malapa kua ntle ga Polokwane, mohlahli o ile a bontšha bana kamoo batho ba lefelo leo ba kilego ba dula le go šoma ka gona. O hlalošitše gabonolo, ka Sepedi gomme ka morago ka Seisemane, gomme a sega gare ga tlhaloso. Kamogelo a nyaka go boloka lentšu la gagwe.

A fihlela mogala gomme ke moka a ema, ka gobane Koko o be a mo lebeletše. "Botšiša pele," Koko a rialo. Kamogelo a botšiša. Mohlahli a nagana gomme a re, "Ee, eupša o swanetše go bolela gore ke mang, le gore se ke sa eng." A bolela ka go hlaka. Ba rekhota kanegelo ya gagwe, go akaretšwa le go sega.

Ge ba boela morago, Tumelo a dira maekerofounu a pampiri a go kgomarela kgauswi le seswantšho. Ka fase ga wona Kamogelo a ngwala, "Botšiša pele. Bolela gore ke mang. Bolela gore ke ka baka la eng." Koko a nagana gore ke karolo ye botse ya letšatši.

Potšišo ya kanegelo: *Modumo o mokopana ofe o ka thušago motho go akanya lefelo la gago?*

Bokgoni: Maekerofounu le modumo

Dira

1. Penta seswantšho sa mohlahli wa musiamo a anela kanegelo ya gagwe. Diriša Oketša ka pela > Oketša modumo. Kgetha Rekhota gomme, ka tumelelo ya motho yo mogolo, o rekhote modumo wa gago wa pampiri ya metsotswana e 3-5 goba molaetša o homotšego wa go theeletša. O se ke wa rekhota motho yo mongwe a sa botšiše.
2. Ka gare ga mohlami wa modumo, goga dibara tša kgetho go dikologa karolo yeo o nyakago go e boloka gomme o diriše Sega.

Theeletša. Rekhota gape ge go nyakega. Ge go se na tumelelo ya maekerofounu, kgetha modumo wa lebokoso la dipuku.

3. Kgetha sprite ya gago. Kgomaretša Letša modumo ka fase ga boloko bja yona bja Thinta-thoma. Kgetha rekhoto ya gago gomme o kgethe 'go fihlela e fela'. Oketša bubble ka morago ga yona.

Khoutu (recipe)

```
SESWANTŠHO SA MOHLAHLI
When this sprite is tapped
Play your chosen sound (until done)
Say in bubble: Ask first. Say who. Say why. for 2 seconds
```

Go šoma bjang

'Go fihlela e fela' e leta gore modumo o fele pele ga boloko bjo bo latelago. 'Gomme o tšwele pele' e dumelela boloko bjo bo latelago go thoma ge modumo o sa letša. Go rekhota ke kgetho ya go hlama; ga go nyakege lentšu la motho ka noši.

Leka

Thinta gatee. Bubble e swanetše go tla ka morago ga ge rekhoto e fedile. Leka 'gomme o tšwele pele' kopong gomme o bapetše. Lekola bophara bja modumo le kgetho ya modumo ge go homologile.

Fetola

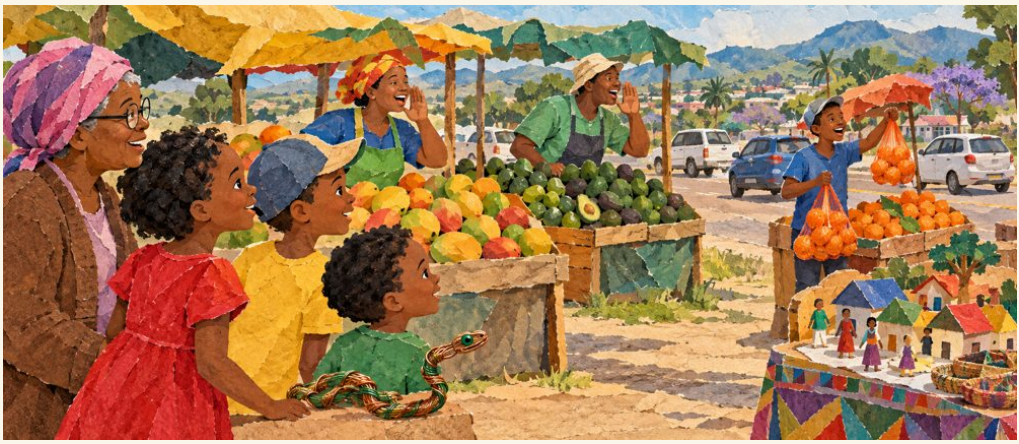
Dira modumo wo o šireletšegilego ka pampiri goba diatla. Leka Modumo wa kgoeledi karolong yeo e kgethilwego, ke moka o bapetše le wa mathomo. O se ke wa goeletša ka maekerofounung.

Hlalosa le go abelana

Dumelela molekane wa gago gore a hlalose gore modumo o oketša eng seo seswantšho se sa se kgonego ka noši. Boloka faele sedirišweng ntle le ge motho yo mogolo a dumela go abelana.

PROJEKE 07

Ke mang yo a bitšago mang



Mabenkeleng a dienywa kgauswi le tsela kua Tzaneen, mang le mang a be a bitša ka nako e tee. Moreki wa mangoro a opela ka monate. Moreki wa diabokado a tšweletša e tee go koloi ye e feto. Mošemane yo a swerego mekotla ya dikgwiri a goeletša ditheko go mang le mang yo a mo lebelelago. Ga go yo a kwelego pitšo e swanago.

Kamogelo a lemoga gore mothekisi yo mongwe le yo mongwe o araba selo se sengwe. Moreki wa mangoro o araba dikoloi tše di fokotšago lebelo. Moreki wa diabokado o araba motho yo a emago. Mošemane o araba tšhelete ye e swerwego moyeng.

Gae ba dira mmaraka wa pampiri teng. Moreki wa mangoro a bitša ge o mo thinta. Moreki wa diabokado a thothomela ge tafola e kgatha. Shato ya thapo e šuthela fela ge letlakala le retollwa. Ditsela tše tharo tša go tsoga.

"Ge ka moka ba be ba araba selo se tee, e be e tla ba lešata fela," Mpho a rialo. Koko a sega. "Ke moka mo fe yo mongwe le yo mongwe kgoro ya gagwe."

Potšišo ya kanegelo: *A motheetši wa gago a ka bapala mokgwa, e sego go o lebelela fela?*

Bokgoni: Ditaki tše di fapanego tša ketsakgato

Dira

1. Penta disprite tše tharo tša dikarolo tša tlhago: mango, abokado le kgwiri go tšwa mabenkeleng a dienywa. Diriša lebokoso la medumo go oketša medumo e meraro e mekopana, goba o rekhote go thinta go šireletšegilego ka tumelelo.

2. Kgetha enywa le lengwe ka nako. E nngwe le e nngwe e fe setaki sa yona sa Thinta-thoma le boloko bja Letša modumo bja modumo wo o kgethilwego wa enywa leo.
3. Diriša 'go fihlela e fela' ka gare ga setaki se sengwe le se sengwe. Beakanya dienywa gore menwana e di thinte ntle le gore di kopane.

Khoutu (recipe)

```
MANGO
When this sprite is tapped
Play sound A (until done)
```

```
ABOKADO LE KGWIRI
Make the same stack on each sprite
Choose a different sound for each
```

Go šoma bjang

Sprite e nngwe le e nngwe e na le khoutu ya yona. Go ema setakeng sa mango ga go emiše setaki sa abokado ka bo yona. Diketsakgato tše di arogilego di ka kopana. Mokgwa wo o hlwekilego o tšwa nakong ya motshameki le khoutu.

Leka

Thinta mango, leta, thinta abokado. Ke moka leka mango le abokado ka pela. Theeletša ge di kopana. Ge enywa le lengwe le lengwe le kwala bjalo, lekola lenaneo la modumo la sprite e nngwe le e nngwe.

Fetola

Thala lepanta la mokgwa: mango, mango, tuduo, kgwiri. Kgopela molekane wa gago gore a le bapale. Ke moka o hlame mokgwa o mongwe.

Hlalosa le go abelana

Fa karata ya mokgwa yeo e bonalago go motho yo a sa kgonego go kwa medumo. Oketša kganya e kopana setakeng se sengwe le se sengwe sa go thinta gore modumo le wona o be le karabo ye e bonalago.

PROJEKE 08

Karata ya poso go tšwa moo dinoka tše pedi di kopanago



Lapa le ile la ya leboa kgole, Mapungubwe, moo Limpopo le Shashe di kopanago. Mohlahli a botša bana gore mmušo o be o gola thabeng moo mengwaga ye ka bago makgolo a seswai e fetilego, le gore batho ba be ba rekišana ka gauta le dikonyana tša galase le lesela le mafelo a kgole. Thaba ye bjale e a šireletšwa. Diswantšho di be di dumelelwa fela moo di dumeletšwego.

Kamogelo o be a tlišitše pampiri le pensele, gomme a thala thaba, dinoka tše pedi le legodimo le legolo kudu. Gae a bea seswantšho sa gagwe godimo ga tafola gomme a tšea seswantšho sa sona. Tafola ka moka ya bonala. Senwelo sa bonala. Lengele la ga Mpho la bonala. Tumelo a mmontšha kamoo a segago, a thala foreime go dikologa fela karolo yeo e nyakegago.

Senwelo sa nyamela. Lengele la nyamela. Seo se šetšego e be e le thaba, dinoka le legodimo. "Ke seswantšho seo o se nyakilego," Koko a rialo. A se bea mothong kgauswi le khalendara.

Potšišo ya kanegelo: *Ke bontšhi bofe bja selo se se tlwaelegilego bjo bo swanetšego go lebelelwa kgauswi?*

Bokgoni: Khamera le go sega diswantšho

Dira

1. Ka tumelelo, Oketša sprite > Khamera gomme o tšee seswantšho sa seswantšho sa thaba le dinoka tše pedi tšeo o di thadilego pampiring ye e se nago selo. Efoga difahlego, dinomoro tša dintlo le dilo tša poraefete. Diswantšho di go dumelela go kgetha seswantšho seo se lego gona seo se amogetšwego.

2. Latela gomme o sege go dikologa karolo yeo o e nyakago, goba o diriše sebopego sa go sega se bonolo. Diriša phimola go tloša dikarolo tšeo o sa di nyakego. Seswantšho sa thaba se se pentilwego ke kgetho ye e lekanago ge khamera e ganetšwa.

3. Kgetha seswantšho sa thaba. Dira setaki sa go thinta se nago le Retolla go ya ka fa letsogong le letona 20, ke moka bubble yeo e hlalosago bontšhi bjo bo tee bjo o bo lebeletšego.

Khoutu (recipe)

```
SESWANTŠHO SA THABA
When this sprite is tapped
Turn clockwise 20 (medium)
Say in bubble: Look at the two rivers! for 3 seconds
```

Go šoma bjang

Seswantšho se ka ba sprite, e sego fela motheo. Go sega go fetola karolo ya seswantšho yeo o e dirišago. Ga go dire gore tshedimošo ya poraefete e šireletšege ge e sa bonala seswantšhong.

Leka

Thinta gatee gomme o lekole gore seswantšho sa thaba se a retologa le go bolela. Thinta gape: go retolla go sengwe go oketša go la mathomo. Se se dirilwe ka maikemišetšo. Ema gomme o se busetše morago ge o nyaka thulaganyo ya mathomo.

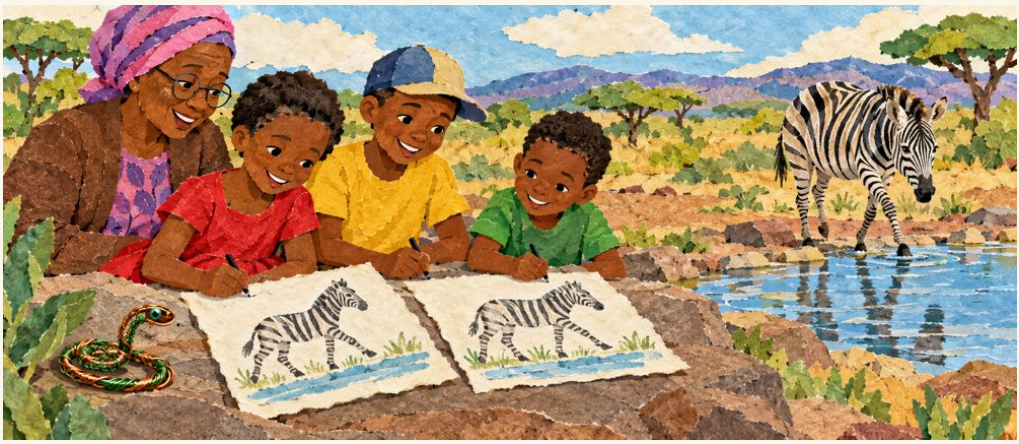
Fetola

Penta seswantšho sa thaba seo se akanyeditšwego kgauswi le seswantšho. Dumelela e nngwe le e nngwe go hlalosa bontšhi bjo bo fapanego. Bolela ka go lebelela le go hlama.

Hlalosa le go abelana

Pele ga go romela selo, hlahloba seswantšho ka moka gomme o theeletše medumo ka moka go nyaka tshedimošo ya poraefete. Motho yo mogolo o dira sephetho sa mafelelo sa go abelana.

Pitsi yeo e sepelago



Polokelong ya diphoofolo ya Polokwane bana ba lebelela dipitsi sediba sa meetse. E nngwe ya gata pele ka leoto la ka pele, ke moka le lengwe. E be e sa akgofe. Mela ya yona e swara seetša sa mathapama.

Mpho a leka go e thala e sepela gomme seswantšho se sengwe le se sengwe se ema. Kamogelo a ba le kgopolo. "Diswantšho tše pedi. Se tee se na le leoto la ka pele le le ka pele. Se sengwe le le ka morago." A di bea ka thoko ga nngwe. Tumelo a di fetoga ka pela, gomme pitsi ya sepela go tshela letlakala.

"Ke ka baka la eng se se šomago?" Mpho a botšiša. Koko, yo a se kego a nea karabo e kopana ge e telele e ka dira, a re leihlo le tlatša seo se tlogetšwego. Go sepela go lokilego go nyaka diswantšho tše pedi fela le motho yo a lebelelago. Bana ba swara pampiri tše pedi bjalo ka dimpho tše nnyane.

Tseleng ya go ya gae pitsi ya šala dihlogong tša bona, e sepela.

Potšišo ya kanegelo: *Ke karolo efe yeo e swanetšego go fetoga, gomme ke efe yeo e swanetšego go dula e eme?*

Bokgoni: Diaparo le animeišene

Dira

1. Diriša OctoStudio 1.1.0 goba e mpsha. Penta sprite ya pitsi. E lokiše go tšwa lenaneong la disprite go bula diaparo tša yona ka Mohlami wa Pente.
2. Kopa seaparo sa mathomo. Fetola fela maemo a leoto la ka pele sa bobedi. Letlalo la onione le ka bontšha seswantšho sa pele gore mmele o dule o beakantšwe.

3. Kgetha pitsi. Ka fase ga Play, oketša Boeletša 4 le Letša diaparo ka gare. E beye gore e letše diaparo tša gago tše pedi tše di bonalago ka lebelo la magareng.

Khoutu (recipe)

```
PITSI
When Play starts
Repeat 4 times:
    Play costumes: 2 (medium)
```

Go šoma bjang

Diaparo ke dibopego tše di fapanego tša sprite e tee. Boloko bja Letša diaparo bo di bontšha ka tatelano. Maemo a mabedi a maoto a boeleditšwe makga a mane a dira animeišene ye bonolo. Se se fapane le go šuthiša sprite ka moka sethaleng.

Leka

Bapala gomme o lebelele fela mmele. Ge o tlola, beakanya diswantšho Mohlaming wa Pente. Ge diaparo di tlhaela, lekola kgatišo ya app le gore diaparo ka bobedi di akareditšwe.

Fetola

Oketša seaparo sa boraro magareng ga leoto la ka pele le la ka morago. Bapetša motšamao. Goba o kgethe emoji yeo e nago le diaparo tše dintši go e na le go thala.

Hlalosa le go abelana

Bontšha diswantšho tše pedi tše di emego pele o bontšha animeišene. Dumelela molekane wa gago go hlalosa kamoo diswantšho tše di emego di fetogago motšamao.

PROJEKE 10

Difahlego tše tharo tša moswi



Magoebaskloof e bitšwa Naga ya Moswi wa Silibera. Ka mosegare wa pele, sethokgwa se be se phuthetšwe ke wona, o boleta le o mošweu. Ka tšatši la gare, moswi wa tloga gomme botala bja tšwa, bo hlabolotšwe le go phadima. Mantšiboa, maru a kgauswi a boa gomme dithaba tša ba gare ga bohlaza le boputsa.

Kamogelo a penta tše tharo ka matlakaleng a arogilego. Mesong e šweu. Mosegare o mošweu o botala. Mantšiboa a magolo a boputsa. Tumelo a di kgomarela ka mothalo mothong.

A bea shato ya thapo letlakaleng le lengwe le le lengwe, e swanago ka mo go feletšego. Ge a fetola letlakala, shato ya dula gomme lefase le le ka morago ga yona la fetoga. Ke ka gona thaba e ikwago ka gona, a rialo. Phoofolo e tee, magodimo a mararo, le sethokgwa seo se sa nyakego go akgofa.

Koko a re lefelo le rotšwe ka Kgoši Makgoba, le gore lentšu kloof le bolela mogwerabaphiri. "Mogwerabaphiri wa Makgoba," a rialo, a tlogela mantšu a dula.

Potšišo ya kanegelo: *Motheetši wa gago o tla tseba bjang gore nako e fetogile?*

Bokgoni: Metheo le ditiragalo

Dira

1. Oketša metheo e mebedi ya lebokoso la dipuku gomme o e rea direkhotong tša gago e le mosegare le mantšiboa. Diriša mafelo a mabedi afe goba afe a a fapanego gabotse; ga a nyake maina a go swana ka app. Motheo wa boraro wa mesong ke kgetho.

2. Penta sprite ya moanegi wa shato ya thapo yeo e akanyeditšwego. Setaki sa yona sa Play se fetoga go ya motheong wa gago wa mathomo ka pela, se leta metsotswana e 3, ke moka se fetogela go wa bobedi.

3. Go moanegi yoo a swanago, oketša setaki se se arogilego sa thomo ya motheo sa motheo wa mantšiboa, se latelwa ke bubble ya polelo.

Khoutu (recipe)

```
MOANEGI: MOTHALO WA NAKO
When Play starts
Switch backdrop to first scene (instant)
Wait 3 seconds
Switch backdrop to evening scene
```

```
MOANEGI: KARABO YA MANTŠIBOA
When evening backdrop starts
Say in bubble: Same forest, new light. for 3 seconds
```

Go šoma bjang

Motheo o ka thoma setaki se sengwe. Kgaetšedi ya semmušo e re ketsakgato ye e thoma magareng ga phetogo. Kgetha diphetogo tša ka pela ge lefelo la gago le nyaka phetogo ye e bonalago gona bjale, gomme o leke nako.

Leka

Bapala gabedi. Lefelo la mathomo le swanetše go boa pele ga phetogo ya mantšiboa. Ge karabo e sa tle, netefatša gore ketsakgato le boloko bja go fetola di kgetha motheo o tee wa bobedi.

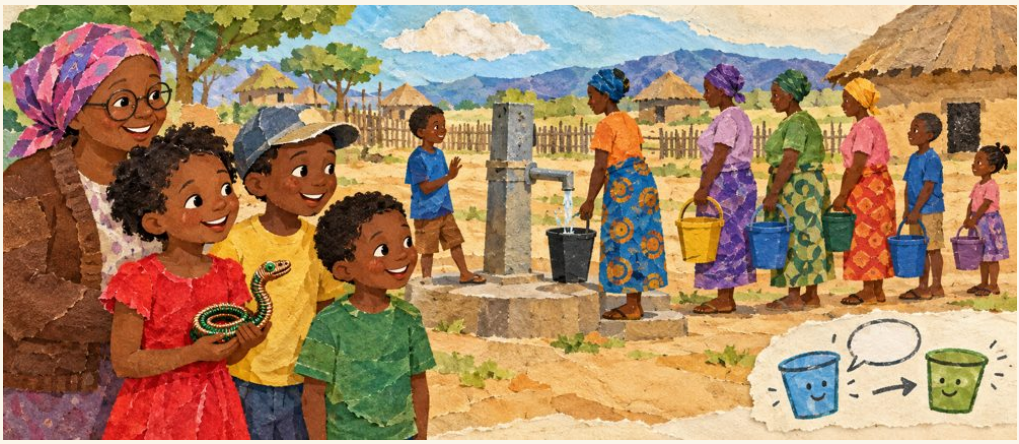
Fetola

Oketša modumo o tee wa lefelo le lengwe le le lengwe. Rekhota polelo ya gago ye e homotšego ya mantšiboa ka tumelelo, goba o tlogele kanegelo e le ya go lebelelwa fela.

Hlalosa le go abelana

Anela kanegelo ka diswantšho tše pedi tša pampiri pele. Ke moka o hlalose gore ketsakgato efe e šuthišetša kanegelo ya gago ya dijithale motsotswaneng wo o latelago.

Nako ya gago sedibeng



Sedibeng sa motse, mothalo wa ditlelo o be o letile. Mang le mang o be a na le felo mo a yago gona. Monna a kgotla leoto. Mosetsana yo monyane a kgokgela. Ge mošemane a kgona go tsena pele, mosadi yo mogolo a emiša monwana o tee gomme a se bolele selo, gomme mošemane a boela morago le tlelo ya gagwe.

"Mang le mang o hwetša meetse," Koko a rialo. "Ka tatelano."

Kamogelo a lebelela kamoo motho yo mongwe le yo mongwe a bolelago lentšu le lekopana go yo a latelago. "Nako ya gago." "Ke a leboga." "Tšwela pele." Ga go lentšu leo le lahlegilego, gomme mothalo wa tšwela pele.

Gae bana ba e dira papadi ya pampiri. Tlelo ye e thadilwego e romela molaetša go tlelo ye e latelago. "Nako ya gago." Ye e latelago e araba, gomme e fetišetša molaetša pele. Ga go yo a bolelago go fihlela molaetša o goroga. Mpho a oketša sebopego sa mafelelo seo se rego fela, "Ke a leboga." Se ile sa ikwa e le tsela ye e swanetšego ya go fetša.

Potšišo ya kanegelo: *Khoutu e ka fetišetša nako ya go bolela bjang ntle le gore mantšu a mabedi a kopane?*

Bokgoni: Melaetša le go tšea dikhutlo

Dira

1. Oketša disprite tše pedi tša baanegi: tlelo le tlelo ya bobedi ye e letetšego ka morago ga yona. Oketša modumo o mokopana wa lebokoso la dipuku go e nngwe le e nngwe, goba o rekhote mela e mebedi e mekopana ya gago ka tumelelo.

2. Go moanegi A, letša modumo wa mathomo 'go fihlela o fela', ke moka Romela molaetša o diriša lenaneo la melaetša. Kgetha molaetša wo o lego gona gomme o ngwale kgetho ya wona ka go lekanela.

3. Go moanegi B, oketša Ge molaetša o amogelwa ka kgetho ye e swanago, ke moka o letše modumo wa bobedi 'go fihlela o fela'. Ye ke molaetša ka gare ga projeke, e sego molaetša wo o romelwago go motho.

Khoutu (recipe)

```
MOANEGI A
When Play starts
Play first line (until done)
Send chosen message to all
```

```
MOANEGI B
When chosen message is received
Play second line (until done)
```

Go šoma bjang

Romela e neelana ka leswao. Boloko bja go amogela ke boloko bja go thoma. Ke molaetša wo o swanago fela wo o thomago setaki sa B. Modumo wa mathomo o fela pele leswao le romelwa, ka gona mantšu ao a beakantšwego a tšea dikhutlo.

Leka

Bapala gatee gomme o theeletše. A e swanetše go fela pele B e thoma. Ge B e homotše, bapetša dikgetho tša melaetša. Ge B e thoma gateetee, lekola gore setaki sa yona se thoma ka Amogela, e sego Play.

Fetola

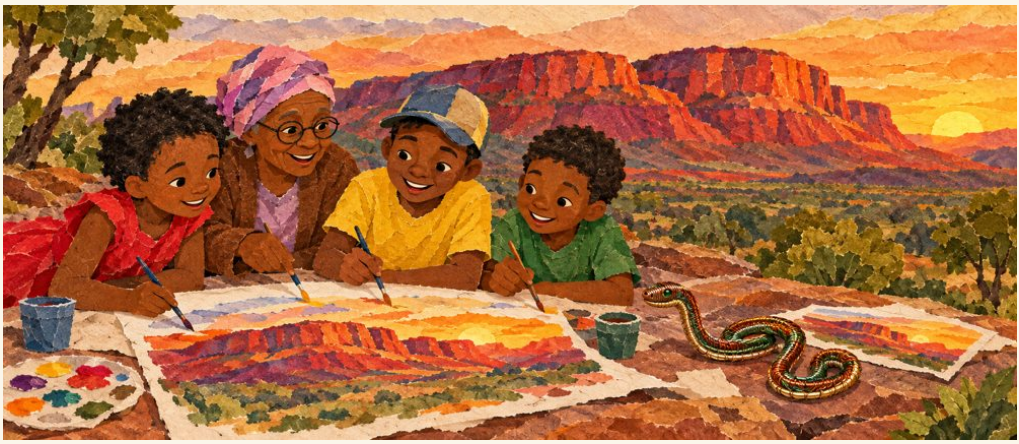
Dumelela B gore e romele molaetša o mongwe wa moanegi wa boraro. Diriša le dikheshene gore tlhaloso e hwetšagale ntle le go theeletša.

Hlalosa le go abelana

Hlalosa phapano magareng ga leswao le la khoutu ya selegae le go romela faele ya projeke go motho yo mongwe.

PROJEKE 12

Lefsika le lehubedu ka morago ga letšatši



Mathapama, mawatle a lefsika la Waterberg a fetoga. Pele a be a le sefotho. Ke moka oranje. Ke moka mmala o mohubedu wo o tseneletšego wo o bego o bonala o tuka ka gare. Kamogelo a swara moya. "Bjalo ka mahlaka," Tumelo a rialo. "Bjalo ka ka gare ga nkgo ya letsopa," Mpho a rialo.

Ba leka go e penta gomme ba šitwa ka ditsela tše tharo. Sefotho se be se le sephara kudu. Oranje e be e le lešata kudu. Koko a šišinya gore ba thome ka mmala wo o lefsifsi gomme ba oketše seetša se sennyane ka godimo, gape le gape. Ba dira bjalo, gomme pampiri ya thoma go phadima.

"Lefsika le tee, mantšiboa a tee," Koko a rialo, "eupša ga go na mmala wo o swanago gabedi."

Kamogelo a dira shato ya thapo ya kopara e lefsifsi e na le mphete ye šoro ya gauta, bjalo ka ge e ka no bago letšatši le e kgomile. A e emiša kgauswi le lefsika seetšeng sa mafelelo, gomme ka motsotsonyana ya bonala bjalo ka ge e be e le gona ka mehla.

Potšišo ya kanegelo: *Maikutlo afe ao o ka a hlamago ka mmala go e na le ka mantšu?*

Bokgoni: Kganya le mmala

Dira

1. Penta sprite ya shato ya thapo motheong wo o lefsifsi wa lebokoso la dipuku, bjalo ka Waterberg ka morago ga letšatši le dikela. E kgethe.

2. Kgomaretša Phadima ka mmala wo mogolo ka fase ga boloko bja Thinta-thoma. Ke moka kgomaretša Bea mmala wa sprite, o kgetha mmala wo o o ratago.
3. Fetola dikgetho tša mmala gomme o bapetše. O se ke wa diriša boloko bja lebone la nnete projekeng ye.

Khoutu (recipe)

```
SHATO YA THAPO YE E PHADIMAGO
When this sprite is tapped
Glow with your chosen warm colour
Set sprite colour to your chosen colour
```

Go šoma bjang

Phadima e dira mothalo wa seetša go dikologa sprite. Bea mmala o fetola mmala wa yona. Ye ke ditlamorago tša sekirini. Ga go nyakege lebone la nnete goba seetša se se phadimago. Go tloša mmala wo o beilwego, kgetha ga go na mmala lenaneong la boloko bjoo.

Leka

Thinta gatee gomme o bapetše shato ya thapo pele le ka morago. Ge dintlha tša yona di nyamela, fetola mmala goba o nolofatše seswantšho. Efoga go phadima ka pela goba diphetogo tše di akgofilego tša mmala.

Fetola

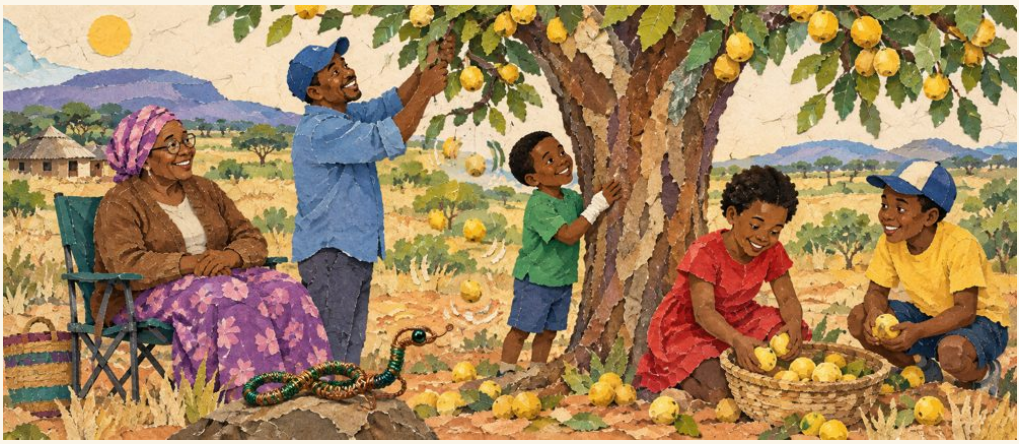
Dira lefelo la bobedi le mmala o homotšego wa go tonya. Botšiša molekane wa gago gore o akanya maikutlo afe, o sa mmotše karabo ya gago pele.

Hlalosa le go abelana

Fa mofuta wa go se phadime go mang le mang yo a sa ikwego gabotse ka ditlamorago tša seetša. Ga go na ngwana yo a nyakago go lebelela sekirini seo se phadimago go ithuta mebala.

PROJEKE 13

Šikinya morula, goba o thinte



Selemong dihlare tša morula di lahla dienywa tša tšona. Ka fase ga se sengwe kgauswi le moagišani wa Koko, fase go be go le serolwana gomme moya o nkga monate. Motho yo mogolo a šikinya lehlare le le fasanyane gomme dienywa tša wela ka lešata. Bana ba lla ka lethabo.

Ke moka Mpho a ngwathela sefahlego. O be a gobetše lengele maloba gomme a sa kgone go šikinya selo. A bea seatla ka fase ga letsogo gomme a leka go bonagala bjalo ka ge go se na bothata.

Kamogelo a lemoga. "Papading ya rena, go šikinya go dira gore shato e sepele," a rialo. "Ke eng ge go thinta le go dira bjalo?"

Ba fetola projeke. Go šikinya goba go thinta, nngwe fela e šuthiša shato ya thapo. Lengele la ga Mpho la tlogela go nyakega. Kamogelo a re papadi ga se ya swanela go ba teko ya gore mmele wa mang o šoma bokaone. Koko, setulong sa gagwe sa go phuthwa, a re fela, "Go lokile. Bjale mang le mang o papading."

Potšišo ya kanegelo: *Projeke e tee e ka amogela ditsela tše di fapanego tša go sepela bjang?*

Bokgoni: Go šikinya le tsela ye nngwe ya go thinta

Dira

1. Kgetha sprite ya shato ya thapo. Dira setaki sa Šikinya-thoma se nago le Tlola 20 le bubble e kopana. Thoma ka kwešišo ye nnyane ya go šikinya.
2. Oketša setaki sa bobedi sprite e tee seo se thomago ge e thintšwa. Se fe go tlola le bubble yeo e swanago gore go šikinya e be kgetho.

3. Swara mogala goba tablet ka diatla ka bobedi godimo ga selo se se bonolo. Dira motšamao o bonolo fela. Go Chromebooks, diriša konope ya go šikinya ya sekirini.

Khoutu (recipe)

```
SHATO YA THAPO: TSELA YA GO ŠIKINYA
When device is gently shaken
Jump 20
Say in bubble: There is a place for you! for 2 seconds
```

```
SHATO YA THAPO: TSELA YA GO THINTA
When this sprite is tapped
Jump 20
Say in bubble: There is a place for you! for 2 seconds
```

Go šoma bjang

Ketsakgato ya sensara e kgokagantšha tiro ya mmele le khoutu. Ketsakgato ya go thinta e ka nea dipoelo tše di swanago. Dilo tše di nyakegago tša hardware le motšamao di a fapana, ka gona tsela ye nngwe ye e kgonegago ke karolo ya sebopego, e sego projeke ya ka fase.

Leka

Leka go šikinya gabonolo gatee, ke moka o thinte gatee. E nngwe le e nngwe e swanetše go dira karabo e tee ye e bonalago. Ge go šikinya go ipoeletša ka bonolo kudu, lokiša kwešišo ya yona goba o diriše tsela ya go thinta fela.

Fetola

Diriša diaparo tše pedi go e na le go tlola ge o na le tšona. Boloka motšamao o le o nnyane go lekana gore o ka o latela.

Hlalosa le go abelana

Botšiša molekane wa gago gore o rata tsela efe. O se ke wa mo gapeletša go šikinya goba go swara sedirišwa ka tsela yeo e sa ikwego gabotse.

PROJEKE 14

Tini ya meetse



Koko a hlwekiša lebato la kitšhi. Tumelo a bea tsela ya khatone, e le bophara mathomong, e kopana bofelong, gomme a bea tini ya meetse godimo ga seleto. Tini e be e swanetše go fihla tafoleng ntle le go tšhologa.

Mpho a segiša seleto kudu gomme meetse a elela ka thoko. Kamogelo a leka ka go se lekane gomme go se sepele selo. Mafelelong a leka go sega go gonnyane, ke moka go gongwe, gomme tini ya šuthela pele. Ge e ama tafola, ka moka ba opa diatla.

Shato ya thapo e be e dutše bofelong, e lebelela. Ya gopotša Kamogelo selo seo Koko a se bolelago gantši. "O rwala meetse ka go nonna ge o nyaka gore a fihle."

Ba dira mofuta wa sekirini, o na le tini ye e thadilwego le tafola ye e thadilwego. Go sega go šuthiša tini. Go ama tafola go dira bubble e re "Go phethilwe." Kamogelo a tlogela tsela e le bophara, ge e ba motho a ka rata dikonope tša metshwari go e na le go sega.

Potšišo ya kanegelo: *Go direga eng ge o gogela ka ditshepedišo tše dinyenyane?*

Bokgoni: Go sega le ketsakgato ya go ama

Dira

1. Penta sprite ya tini ya meetse le sprite ye kgolo ye e pentilwego ya sepheo sa tafola. Bea sepheo kgole le tini ya mathomo gore di se ke tša ama. Dira setaki sa Play godimo ga tini seo se yago felong la mathomo leo le kgethilwego.

2. Ka morago ga go busetša, diriša Sega go tšwa go Dibolokwa tše dingwe ka gare ga Ka mehla. Gantši e tla le boloko bja Ka mehla. Bea ditsela ka moka tše nne le lebelo le le nnyane. Swara sedirišwa ka kgonego ya sona ya nnete pele ga Play.

3. Godimo ga tini, oketša ketsakgato ya go thoma go ama yeo e lebišitšwego sepheong. Oketša bubble ya moketeko ke moka Emiša ditaki ka moka. Go Chromebook diriša kgatišo ya yona ya go sega ya sekirini.

Khoutu (recipe)

```
TINI: GO GOGELA
When Play starts
Go to chosen start position (instant)
Forever:
    Tilt to move (four directions, slow)
```

```
TINI: GO FIHLA
When tin touches goal
Say in bubble: Done! for 2 seconds
Stop all scripts
```

Go šoma bjang

Ka mehla e tšwela pele go lekola go sega ge papadi e šoma. Ketsakgato ya go ama e nyaka disprite tše pedi. Emiša tšohle e fediša ditaki tše di šomago tša projeke; konope ya Ema le yona e ka e emiša nako efe goba efe.

Leka

Boloka disprite ka bobedi kgole le mekgwa. Gogela ka go nonna sepheong. Lekola gore papadi e a ema. Bapala gape go busetša tini. Ge sedirišwa se se na le go sega go go dirišegago, diriša ditaolo tša Chromebook goba tsela ye nngwe ya ka fase.

Fetola

Tsela ye nngwe: Emiša karolo ya Ka mehla/go sega ka disprite tše nne tša metshwari. Godimo ga tini dira ditaki tše nne tša sepheo sa go thinta, e nngwe le e nngwe e e šuthiša dikarolo tše 10 ka tlhoko ya metshwari yeo.

Hlalosa le go abelana

Thala tsela pele o bapala. Dira tsela e bophara go e na le maze a thata. O se ke wa sega Chromebook ka mmele.

PROJEKE 15

Dilatelo tše tharo tša pitša



Ka nako ya yona, malapa a kgoboketša masonja, difene tša mophane, dihlareng tša mophane. Koko a iša bana sethokgweng sa mophane le lapa la moagišani, moo mang le mang a šomilego ka kelohlokomelo, gomme ga go selo seo se ilego sa tšewa gabedi.

"Dilatelo tše tharo di tla lekana pitša ya bošegong bja lehono," Koko a rialo. "Bala ge o ntše o sepela." Mpho a bala ka go hlaboša lentšu. Tumelo a boloka palo thoko ga tlelo. Letšhwao le tee go latelo le lengwe le le lengwe.

Gae Kamogelo a nyaka gore palo e namele ka bo yona. A thala dithokwana tše tharo tša masonja pampiring le tlelo kgoreng. Nako le nako ge a thinta thokwana, ya tsena tlelong, gomme lebokoso le lennyane la oketša le tee. Thokwana yeo e thintšwego e be e sa kgone go thintšwa gape.

Ge tše tharo ka moka di be di le ka gare, lebokoso la re tharo. Koko a emiša tlelo ya nnete go ikwa boima. Kamogelo a ngwala palo seatleng sa gagwe ge e ba a e lebala, gomme ke moka, ka gobane a rata kamoo e bonagalago, a e tlogela moo go fihlela nako ya go hlapa.

Potšišo ya kanegelo: *Palo e ka gopola tiro bjang ntle le go e bala gabedi?*

Bokgoni: Dibolokwa tše di fetogago le dintlha

Dira

1. Penta disprite tše tharo tše di bonalago tša dilatelo tša masonja motheong wa kitšhi. Diriša diswantšho, e sego dijo tša nnete. Go Dibolokwa tše dingwe, bula lenaneo la bolokwa bjo bo fetogago

gomme o diriše + go dira 'dilatelo'. Maina a dibolokwa a ka se tlošwe bjale, ka gona kgetha ka kelohlokomelo.

2. Go latelo A, aga Play > Bea dilatelo go 0 > Bontšha ka 100%. Go dilatelo B le C diriša Play > Bontšha ka 100%. Dibolokwa di bontšhwa ka tlwaelo; diriša Bontšha bolokwa bjo bo fetogago ge go nyakega.

3. Go latelo le lengwe le le lengwe, oketša Thinta sprite ye > Fetola dilatelo ka 1 > Uta ka pela. Ge e phihlilwe, ga e sa kgone go thintšwa gape. Gatiša Play gomme o leta gore mathomo a fele pele ga go kgoboketša.

Khoutu (recipe)

```
LATELO A: MATHOMO
When Play starts
Set variable handfuls to 0
Show at 100% (instant)
```

```
LATELO LE LENGWE LE LENGWE: GO BALA
When this sprite is tapped
Change variable handfuls by 1
Hide (instant)
```

Go šoma bjang

Bolokwa bjo bo fetogago ke palo ye e nago le leina yeo khoutu ya gago e ka e ntšhafatšago. Bea e dira lefela le lefsa. Fetola e oketša le tee. Uta e thibela gore latelo le tee le se kgoboketšwe gape. B le C di nyaka ditaki tša tšona tša Bontšha-ka-Play, e sego go busetša palo gape.

Leka

Ka morago ga Play, thinta latelo le lengwe le le lengwe gatee. Solofela 1, 2, 3. Thinta felo mo go se nago selo moo latelo le bego le le gona: ga se a swanela go oketša gape. Thoma gape gomme o solofele 0 le dilatelo ka moka di bonala. Thinta ka go nonna ge o leka.

Fetola

Dira dilatelo tše hlano gomme o fetole sepheo. Boloka yo a busetšago a le tee gore go be bonolo go mo hwetša. Leka go thinta ka pela sedirišweng sa nnete pele o se diriša bjalo ka papadi ya phapoši ya borutelo.

Hlalosa le go abelana

Botša molekane wa gago gore ke ka baka la eng 'fetola dilatelo ka 0' e ka se busetšego dintlha tša maloba.

PROJEKE 16

Koko o lekola tlelo



Mantšiboa ao Koko a ema kgauswi le setofo a na le pitša ye kgolo le leswana la kota. "A dilatelo tše tharo ka moka di ka gare?" a botšiša, a sa retologe. Bana ba lebelelana. Ga go yo a bego a bolela.

Kamogelo a bula projeke yeo a e dirilego. A oketša Koko bjalo ka sebopego se sefsa gomme a mo fa molao. Ge Koko a thintšwa, o lekola palo. Ge e feta tše pedi, o re, "Ke a leboga! The pot is full." Ge go se bjalo, o re, "Find all three handfuls first."

Tumelo a thinta Koko ka morago ga latelo le tee. "Find all three handfuls first," ga bolela Koko yo a thadilwego. Ka morago ga tše pedi, bjalo ka mo. Ka morago ga tše tharo, a re "Ke a leboga! The pot is full," gomme Koko wa nnete a sega gomme a opa diatla.

"O ntirile motho yo a lokilego kudu," Koko a rialo. "Nka se e bolele bokaone." A retologela setofong gomme kitšhi ka moka ya nkga dijo tša mantšiboa.

Potšišo ya kanegelo: *Projeke ya gago e swanetše go dira eng ge sepheo se sa fihlelwe?*

Bokgoni: Maemo le sepheo

Dira

1. Kopa Projeke 15. Oketša sprite ya Koko, o boloke khoutu ya gagwe e arogile le ya dilatelo.
2. Go Koko, kgomaretša boloko bja Ge ka fase ga Thinta sprite ye. Tsenya morepoti wa bolokwa bjo bo fetogago bja dilatelo lekgobong la mathomo la palo. Kgetha go feta gomme o bee 2 lekgobong le lengwe.

3. Ka gare ga Ge bea bubble 'Ke a leboga! The pot is full.' ka metsotswana e 2. Thinta tlaleletšo ya boloko bja Ge go oketša Ge go se bjalo; ka gare ga Ge go se bjalo bea 'Find all three handfuls first.' ka metsotswana e 2.

Khoutu (recipe)

```
KOKO
When this sprite is tapped
If handfuls > 2:
    Say: Ke a leboga! The pot is full. for 2 seconds
Else:
    Say: Find all three handfuls first. for 2 seconds
```

Go šoma bjang

Ge e lekola maemo gatee, ge go thinta go e fihlela. Ka ntlha e tee ya latelo le lengwe le le lengwe, dilatelo di feta 2 di bolela gore tše tharo ka moka di kgobokeditšwe. Ge go se bjalo ke tsela ye nngwe. Tlhahlobo ye e tsamaišwago ke ketsakgato ga e nyake Ka mehla ka gobane go thinta gongwe le gongwe go dira tlhahlobo ye mpsha.

Leka

Thinta Koko ge dilatelo e le 0, 1, 2 le 3. Ke ya mafelelo fela yeo e swanetšego go nea molaetša wa teboho. Thoma gape gomme o lekole gore Koko o nea molaetša wa go se fele gape.

Fetola

Ge go na le dilatelo tše hlano, fetola maemo go ba dilatelo > 4. Bolela gore ke ka baka la eng go fetola seswantšho fela go ka tlogela molao wa kgale e le phošo.

Hlalosa le go abelana

Dira seswantšho sa pampiri sa ditsela tše pedi: sepheo se fihletšwe / ga go yo. Fa maikutlo a a thušago ditseleng ka bobedi.

PROJEKE 17

Leru le tla ba legolo bjang



Mantšiboa a go fiša ka fase ga mohlare wa Koko, bana ba bo malome ba kgopela kanegelo. "Ba re," Koko a thoma, e leng ka fao a anelago tša kgale, "gore kgalekgale dikgošikgadi tša Modjadji tša Balobedu di be di bolelwa gore di bitša pula. Ke seo batho ba kgale ba se bolelago."

Tumelo a lebelela legodimo. E be e se na selo. Maru a mmalwa a manyane a be a dutše kgole.

"A re e dire papadi," Kamogelo a rialo. "Leru la pula le tla ba legolo bjang?" Mpho a tliša dae. A thala leru la pula la pampiri magareng ga letlakala. Nako le nako ge a le thinta, la ba le bogolo bjo bo fapanego. Ka dinako tše dingwe e be e le sefokotšana. Ka dinako tše dingwe le be le batla le tlala letlakala. Gabedi ka go latelana la ba le bogolo bjo bo swanago, gomme ga go yo a ilego a ngongorega.

Koko a re e be e le kanegelo fela, gomme kanegelo ye botse ga e nyake go dumelwa gore e rategwe. Bana ba tlogela leru magareng gomme ba dumelela dae gore e kgethe.

Potšišo ya kanegelo: *A kgetho ye e sa beakanywago e ka boeletša, gomme e sa dutše e sa beakanywa?*

Bokgoni: Dipalo tše di sa beakanywago

Dira

1. Oketša sprite ya leru la pula yeo o e pentago. Ketsakgatong ya go thinta kgomaretša Bea bogolo. Lekgobong la yona la palo, kgomaretša morepoti wa Palo ye e sa beakanywago go tšwa go Dibolokwa tše dingwe.

2. Bea bophara bja go se beakanye go tloga go 40 go fihla go 100. Kgetha diphetogo tša bogolo tša ka pela gore dipoelo di bapetšwe ka bonolo.

3. Boloka leru kgauswi le magareng a sethaleng. Dipalo tše dikgolo tša bogolo di ka nyaka seswantšho sa mathomo se senyane.

Khoutu (recipe)

```
LERU LA PULA
When this sprite is tapped
Set size to random number from 40 to 100 (%)
```

Go šoma bjang

Morepoti wo o kgokologilego o nea palo go boloko bjo bongwe. Ga o dule o le noši bjalo ka tiro. Bophara bo bea boleng bjo bo kgonegago. Go thinta gabedi go ka nea boleng bjo bo swanago; go se beakanywe ga go bolele gore ka mehla go fapana.

Leka

Thinta makga a mahlano gomme o bapetše bogolo. E nngwe le e nngwe e swanetše go dula magareng ga dintlha tša gago tšeo o di kgethilego. Ge go se fetoge selo, lekola gore morepoti o kgomaretšwe ka gare ga lekgoba la bogolo, e sego o robetše o lokologile kgauswi.

Fetola

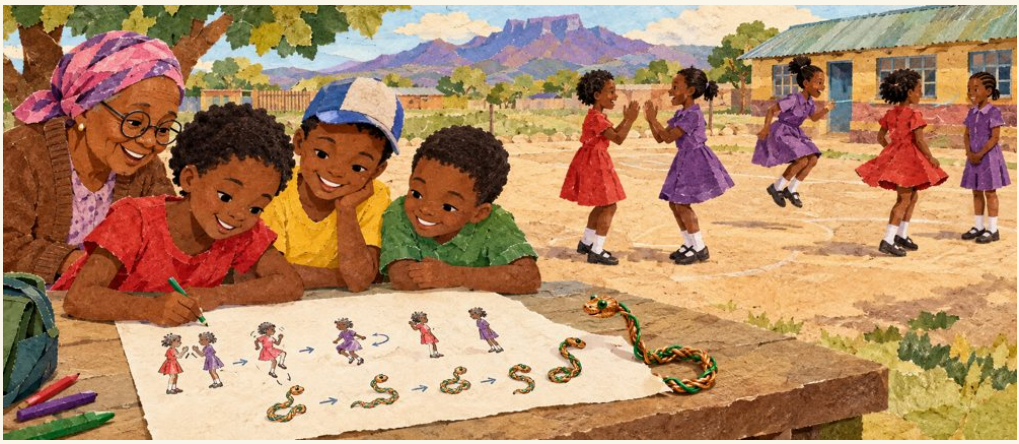
Leka 50-60 gomme o bapetše le 40-100. Ke bophara bofe bjo bo dirago gore diphetogo di be bonolo go di lemoga? Ngwala tebelelopele pele.

Hlalosa le go abelana

Kgokagantšha papadi ya dae ya nnete le kgetho ya palo ya sekirini. Ka bobedi di nyaka sethaopo se se hlakilego sa dipoelo tše di kgonegago.

PROJEKE 18

Go opa diatla ga Koko



Lebaleng la sekolo, banenyana ba be ba na le papadi ya go opa diatla. Tlola, retolla ka botlalo, leta molekane wa gago. E be e sepela ka mokgwa woo o dirilego gore mang le mang a sege. Kamogelo a e lebelela makga a lesome le bobedi. Nako le nako ge ba thoma, yo mongwe o be a swanetše go goeletša kgato e nngwe le e nngwe.

Gae tafoleng, a dira gore shato ya thapo e dire bjalo: tlola, retolla ka botlalo, leta. Nako le nako ge a e nyaka, o be a swanetše go aga dikgato ka moka tše tharo gape. Tumelo a re, "E fe leina."

Ba e bitša "go opa diatla ga Koko," ka gobane Koko o ba rutile mofuta wa mathomo mengwaga e mentši e fetilego. Bjale leina le tee le swere dikgato tše tharo. Ge Kamogelo a bitša "go opa diatla ga Koko," shato ya dira tšohle ka noši.

Koko a theeletša tlhaloso, a dumela, gomme a re o be a tseba ka mehla gore leina le ka rwala selo se setelele ka gare ga lona. Ke moka a botšiša gore ke mang yo a nyakago tee.

Potšišo ya kanegelo: *Ke tatelano efe ye e boeletšwago ye e swanetšego go ba le leina la yona?*

Bokgoni: Dibolokwa tše di ikgethilego

Dira

1. Oketša sprite ya shato ya thapo. Go Dibolokwa tše dingwe, dira boloko bjo bo ikgethilego bjo bo bitšwago 'go opa diatla ga Koko'. Hwetša tlhaloso ya bjona gomme o kgomaretše Tlola 20, Retolla 180 go ya ka fa letsogong le letona, ke moka Leta 1 ka fase ga yona.

2. Setakeng se segolo sa shato, bea Boeletša 2 ka fase ga Play. Bea boloko bjo bo mpsha bja go opa diatla ga Koko ka gare ga Boeletša.
3. O se ke wa bea tlhaloso ka gare ga setaki se segolo. E hlalosa seo boloko bjo bo mpsha bo se dirago ge bo bitšwa.

Khoutu (recipe)

```
SHATO YA THAPO: GO HLALOŠA
Define Koko's clap
Jump 20
Turn clockwise 180
Wait 1 second
```

```
SHATO YA THAPO: GO DIRIŠA
When Play starts
Repeat 2 times:
    Koko's clap
```

Go šoma bjang

Boloko bjo bo ikgethilego bo nea tatelano leina. Go bo bitša go dira ditiro tšeo di lego tlhalošong ya bjona. Go OctoStudio dibolokwa tše di ikgethilego ke tša sprite yeo e di hlamilego. Ga di bonale ka othomathiki go disprite tše dingwe ka moka.

Leka

Bapala: solofela go tlola gabedi le go retolla gabedi ka go ema. Fetola go tlola tlhalošong gomme o bapale gape. Ditirišo ka bobedi di swanetše go fetoga ka gobane di bitša tlhaloso e tee.

Fetola

Dira kgato ya bobedi ye e filwego leina ka motšamao wo o fapanego. Diriša maina a mabedi ka tatelano ye e anelago mathomo, magareng le bofelo.

Hlalosa le go abelana

Dumelela molekane wa gago a hlalose setaki se segolo a sa bale tiro e nngwe le e nngwe. Ke moka a bule tlhaloso go bona gore e šoma bjang.

PROJEKE 19

Kanegelo ya motho yo a lego kgole



E be e le beke ya mafelelo ya Kamogelo kua Polokwane. Ka Mokibelo bese e be e tla mo iša gae Johannesburg. A nyaka gore tsala ya gagwe Palesa a bone tšohle. Eupša Palesa o be a se na OctoStudio, gomme mogala wa gagwe o be o tlhaela sebaka ka mehla.

Koko a botšiša potšišo ye e lokilego. "A o nyaka go lebelela, goba o nyaka go e fetola?" Kamogelo a nagana. Ge Palesa a be a nyaka go bona fela, rekhoto e kopana e be e tla lekana. Ge a nyaka go thinta, go šikinya le go fetola mantšu a shato, o be a tla nyaka faele ya nnete ya projeke le app.

Ba dira ka bobedi. Vidio e nnyane gore e romelwe bošegong bjoo. Le faele ya projeke, e bolokilwe gabotse, gore Palesa a e bule ge a ikemišeditše. Kamogelo a ngwala lengwalo ka seatla gomme a le tsenya bukeng ya gagwe. "Ye ke ya go lebelela. Ye ke ya go fetola."

Potšišo ya kanegelo: *Motheetši wa gago o nyaka go lebelela, goba go bula le go fetola khoutu?*

Bokgoni: Vidio, GIF le faele ya projeke

Dira

1. Bula kopi ya Projeke 9 goba projeke ye nngwe ye kopana ye e fedilego. Sekaseka seswantšho se sengwe le se sengwe, lentšu le modumo le motho yo mogolo. Tloša tshedimošo ya poraefete pele ga go romela.

2. Bula Abelana > Rekhota Vidio. Thoma go rekhota gomme o dire projeke o diriša ditaolo tša sekirini. Ema ka fase ga metsotswana e 30. Diriša kgetho ya go boloka ya sedirišwa ge e gona.
3. Leka Abelana > Dira GIF ya loop e kopana e homotšego, go fihla go metsotswana e 15. Ke moka Abelana > Romela Faele ya Projeke gomme o boloke kopi ya .octostudio selegaeng. O ga o hloke go romela le ge e le efe ya tše go motho yo mongwe.

Khoutu (recipe)

```
TIRAGATŠO E KOPANA YA GO REKHOTA
Use your completed greeter or story project
Press Play and show one clear beginning and ending
Keep video under 30 seconds; GIF under 15
```

Go šoma bjang

Vidio e bontšha tiragatšo e tee ye e rekhotilwego gomme e ka akaretša modumo. GIF ke loop e kopana ye e bonalago ye e se nago modumo. Faele ya projeke e boloka projeke ye e ka lokišwago, ye e dirišanago gomme e nyaka OctoStudio gore e bulwe. Rekhoto ga e boloke go thinta goba go sega ga nako ya nnete.

Leka

Bula rekhoto ye e bolokilwego gomme o lekole tiragatšo ka moka. Bula faele ya projeke ka OctoStudio gomme o lekole dibolokwa tša yona. Ge letlapa la go abelana le fapane, kgopela motho yo mogolo go nyaka tsela ya go boloka faele selegaeng go e na le go e bea inthaneteng.

Fetola

Beakanya GIF ye e homotšego le vidio ye e anetšwego ya kgopolo e tee. Bapetša gore ke tlhaloso efe yeo e šalago ntle le modumo.

Hlalosa le go abelana

Boloka kopi pele o tloša app goba go phimola data ya app. Go abelana inthaneteng go ka diriša data gomme go nyaka tlhahlobo ya babogedi le poraefete ya motho yo mogolo. Ga go na ngwana yo a swanetšego go gatiša mošomo wa gagwe go fetša puku ye.

PROJEKE 20

Seo re se boelago le sona



Mantšiboa a mafelelo ba boela Magoebaskloof gatee gape. Moswi o be o thomile go kgobokana mogwerabaphiri. Koko a dula lefsikeng le le sephara gomme bana ba dula go mo dikologa, bjalo ka ge ba dira bjale ka mehla.

Kamogelo a bula puku ya gagwe. E be e tletše. Shato kgorong. Tsela ye e nago le tatelano. Mmete wo o retologago. Mokwa wo o tšeago nako. Kgwara yeo e sa tsogego. Lentšu la mohlahli, leo le kopilwego pele. Mmaraka, thaba magareng ga dinoka tše pedi, pitsi, magodimo a mararo. Mothalo wa sedibeng, lefsika le lehubedu, tini ya meetse. Dilatelo tše tharo le pitša e tletšego. Leru la bogolo bofe goba bofe. Go opa diatla ga Koko.

Ka thoko ga tsela go be go na le lehlaka la lebenkele la dimonate. Tumelo a le tšea gomme a le tsenya kgetsaneng. Kamogelo a mmona gomme a tseba gore projeke ya gagwe ye mpsha e tla re eng.

E dirišitše tšohle. Moetapele wa shato ya thapo yo a go amogelago, lehlaka leo o ka le thintšago go thuša, le bofelo bjo bo fapanego ge o dira bjalo. Ge e fedile, Koko a bolela mantšu a kgale: "Motho ke motho ka batho." Kamogelo a thinta shato. "Thobela," ya re. Ka moka ba araba, gomme moswi wa tšwela pele go tla.

Potšišo ya kanegelo: *Phetogo efe yeo o ratago gore motho a e akanye setšhabeng sa gago?*

Bokgoni: Itlhamele kanegelo ya gago ye e dirišanago

Dira

1. Beakanya kanegelo ya tlhokomelo ya lefelo ka nako tše tharo pampiring: tikologo, selo seo motheetši a ka se dirago, le bofelo. Kgetha diswantšho tša mathomo goba disprite tša lebokoso la dipuku, metheo e mebedi ye e amogetšwego le modumo o tee o kopana wa kgetho.
2. Aga sedirišwa sa mathomo se se ka fase ka sprite ya Moetapele wa Shato ya Thapo le sprite ya Lehlaka, lehlaka la lebenkele la dimonate tseleng. Kgetha molaetša o tee wa ka gare wa 'ready' le o mongwe wa 'ending'; di swanetše go fapana. Boloka Moetapele a bonala metheong ka bobedi.
3. Diriša dikgetho tše di swanago tša go romela/go amogela. Oketša mantšu a gago le karolo e tee go tšwa projekeng ya pele: diaparo, boloko bjo bo ikgethilego, mmala goba go boeletša go go balwago. Boloka karolo yeo e le kopana go lekana gore e leke.

Khoutu (recipe)

```
MOETAPELE: GO AMOGELA
When Play starts
Switch to welcome backdrop (instant)
Say: Tap the scrap to help. for 3 seconds
Send READY message to all
```

```
LEHLAKA: GO BUSETŠA LE GO BONTŠHA
Stack 1: When Play starts > Hide (instant)
Stack 2: When READY message received > Show (instant)
```

```
LEHLAKA: GO THINTA GO FETŠA
When this sprite tapped
Hide (instant)
Send ENDING message to all
```

```
MOETAPELE: BOFELO
When ENDING message received
Switch to second backdrop (instant)
Say: Together we can care for our home. for 3 seconds
```

Go šoma bjang

Lehlaka le na le ditaki tše tharo tše di arogilego: Play > Uta; Amogela READY > Bontšha; Thinta Lehlaka > Uta > Romela ENDING. Setaki sa Play se a le busetša. Melaetša e beakanya kanegelo go disprite ka moka. Diriša kgetho ye e swanago ya molaetša go romela le go amogela ka bobedi.

Leka

Bapala go tloga mathomong: lehlaka le phihlilwe ge go amogelwa, le a bonala ka morago ga READY, ke moka go thinta gatee go fetola lefelo

gomme go le uta. Thoma gape gomme o boeletše. Lekola dikgetho tša melaetša le dibolokwa tša go busetša pele ga go oketša dikarolo tše dingwe.

Fetola

Dira sepheo e be sa gago, e sego fela papadi ya matlakala. Dumelela tsala gore e leke ntle le go e kaela. Lebelela gore a ema kae, botšiša gore ke eng seo se mo gakantšhitšego, gomme o fetole selo se tee.

Hlalosa le go abelana

Boloka kopi ye e ka lokišwago. Bontšha kanegelo, hlalosa ketsakgato e tee le loop e tee goba kgetho, gomme o bolele phetogo yeo o e dirilego ka morago ga go leka. Kgetha go romela go go šireletšegilego fela ka morago ga tlhahlobo ya motho yo mogolo.

Dira gore e šome go yo mongwe

Leka, lebelela, lokiša

Bapala go tloga mathomong: lehlaka le phihlilwe ge go amogelwa, le a bonala ka morago ga READY, ke moka go thinta gatee go fetola lefelo gomme go le uta. Thoma gape gomme o boeletše. Lekola dikgetho tša melaetša le dibolokwa tša go busetša pele ga go oketša dikarolo tše dingwe.

Dira e be ya gago

Dira sepheo e be sa gago, e sego fela papadi ya matlakala. Dumelela tsala gore e leke ntle le go e kaela. Lebelela gore a ema kae, botšiša gore ke eng seo se mo gakantšhitšego, gomme o fetole selo se tee.

Botša yo mongwe

Boloka kopi ye e ka lokišwago. Bontšha kanegelo, hlalosa ketsakgato e tee le loop e tee goba kgetho, gomme o bolele phetogo yeo o e dirilego ka morago ga go leka. Kgetha go romela go go šireletšegilego fela ka morago ga tlhahlobo ya motho yo mogolo.

Go naganišiša ga moitiri wa gago

Kanegelo ya ka e mabapi le ___. Motheetši wa ka a ka ___. Ketsakgato ye e thomago projeke ya ka ke ___. Khoutu ya ka e gopola goba e boeletša ___. Mathomong ke be ke solofetše ___. Ka morago ga go leka, ke fetotše ___. Kgopolo ya ka ye e latelago ke ___.

Tlhahlobo ya molekane ya botho

Botšiša: Ke motsotso ofe wo o dulago le wena? A go be go na le karolo yeo o sa kgonego go e hwetša goba go e diriša? A dikheshene, sprite ye kgolo goba tsela ye nngwe ya go tsenya di ka thuša? Fa tshišinyo e tee ye e kgethegilego, e sego dintlha tša gore ke mang yo a akgofilego.

Ge disedirišwa di fapana

Ga go na ngwana yo a tlogelwago ka ntle ga projeke

Khamera e ganetšwa? Penta selo. Maekerofounu e ganetšwa? Diriša modumo wa lebokoso la dipuku le kheshene. Ga go na go sega? Diriša tsela ye nngwe ya disprite tša metshwari Projekeng 14. Ga go na go šikinya? Diriša setaki sa go thinta Projekeng 13. Go Chromebooks diriša taolo ya go sega ya sekirini le dikonope tša go šikinya goba makenete; o se ke wa šikinya laptop ka mmele.

Makenete, go thothomela le lebone ke dikgetho

Kgaetšedi ya semmušo e na le boloko bja go thoma ka makenete, eupša disedirišwa tše dingwe tša Android ga di bo thekge. E na le le go thothomela: ke mogala wo o nago le haptic feedback fela wo o thothomelago, disedirišwa tše dingwe di letša modumo wa buzz. Boloko bja flashlight bo nyaka hardware ye e swanetšego gomme bo ka phadimiša sethale go e na le moo. Ga go na selo seo se nyakegago puku ye. O se ke wa diriša makenete a maatla, seetša se se phadimago goba didirišwa tše bohlokwa kopanong ya bana.

Beam ga se molaetša wa poraefete

Beam e romela leswao la Bluetooth go disedirišwa tša OctoStudio tše di lego kgauswi; e fapane le Romela/Amogela ka gare ga projeke yeo e dirišitšwego go Diprojeke 11 le 20. Kgaetšedi e re di-phone le di-tablet di ka romela; Chromebooks di ka amogela eupša ga di kgone go romela. Dipeakanyo tša Bluetooth le ditumelelo di a e ama. Tsela e thuša go aroganya dihlopha eupša ga se kano ya sephiri.

Pontšho ya motho yo mogolo fela, ke kgetho: mogaleng o tee goba tablet, thoma ka Thinta > Beam leswao leo le kgethilwego tseleng ye e kgethilwego. Sedirišweng se se kgauswi diriša Ge beam eo e amogelwa > bubble ya polelo. Swantšha leswao le tsela, bula Bluetooth ka tumelelo, gomme o leke kgole ya kgauswi. O se ke wa akaretša data ya poraefete. Ge e šitwa, boela tirong ya molaetša ka gare ga projeke.

Tiro ya hardware le go tsena di swanetše go lekolwa disedirišweng tšeo di dirišwago tota. Ga go na tiro yeo e ikanyago Bluetooth, makenete, lebone goba go thothomela. Go šoma ntle le inthanete ga go bolele gore sedirišwa se sengwe le se sengwe se na le sensara e nngwe le e nngwe.

Go dira ga mathomo ga moithuti yo a thušago

Pele ga go ruta ka moka tše 20

- Lekola dikarolo tša nnete tša disedirišwa, dikgatišo tša app, puo, sebaka le ditumelelo. Diriša sedirišwa se tee sa Android goba iOS ka babedi ge go kgonega; akaretša diphetogo tša Chromebook ge e le dilo tša gago.
- Dira Diprojeke 1-5 ka bowena ka diswantšho le media ya lebokoso la dipuku. Netefatša dikgetho tša sepheo sa go thinta, go busetša ga bogolo, boemo bja loop le tiro ya uta/bontšha. Tšea diswantšho tša sekirini tša nnete fela ge puo le kgatišo ye e kgethilwego e lokišitšwe.
- Leka media le diaparo Diprojekeng 6-9. Lekola ditsela tše dingwe ge ditumelelo di ganetšwa. Leka nako ya moanegi, phetogo ya lefelo le melaetša ye e swanago ya ka gare ga projeke Diprojekeng 10-11.
- Lekola go šikinya/go sega le go kgotlhana Diprojekeng 13-14. Leka tsela ye nngwe ya konope ya metshwari le motho yo a sa kgonego goba a sa nyakego go sega. Netefatša gore go fihla go ema ka botshepegi.
- Leka go kgoboketša dintlha ka go nonna le ka pela Diprojekeng 15-16. Netefatša ntlha e tee ya latelo le lengwe le le lengwe leo le thadilwego, go thoma gape go ya go zero le maikutlo a nepagetšego a sebopego. Ge go thinta ka pela go ka bala gabedi sedirišweng, tšea se e le phošo yeo e swanetšego go lokišwa pele ga go ruta.
- Leka tsela ya kopi/go bula selegaeng le diromelo ka moka tše tharo Projekeng 19. Ke moka leka Projeke 20 go tloga mathomong a hlwekilego, go akaretša go thoma gape makga a mantši, go thinta ka pela le melaetša ye e fetotšwego.

Ruta ka dikgato tše kopana tša go dira

Bala kanegelo ka go hlaboša lentšu ntle le go e aroganya go bolela maina a dibolokwa. Botšiša potšišo e tee ye e nago le bohlokwa. Bontšha tiro e tee ye mpsha ya sekirini, ke moka o dumelele babedi gore ba e leke. Fa nako ya go fetola seswantšho le mantšu. Fetša ka go bontšha, go hlalosa le go boloka kopi, e sego lebelo la go fetša. Tše ke diprojeke, e sego lenaneo le le tiišitšwego la projeke e tee ka beke.

Pele ga go ntšha: lekola kwešišo ya bana, ditokelo tša media, tiriso ye e šireletšegilego le go balega ga kgatišo. O se ke wa bolela gore puku ye e amogetšwe ke DBE goba gore e na le mmapa wo o feletšego wa CAPS.

Bona seo moithuti a se kwešišago

Theeletša ditlhaloso, e sego fela dipoelo

Diriša ditlhaloso tše nne: "Ke e lekile ka thušo", "Ke a e kgona", "Ke ka e hlalosa", "Ke ka e fetolela yo mongwe". Ke ditlhotlheletšo tša thekgo, e sego maina ao a kgomarelwago ngwana. Ngwala fela seo se nyakegago phapošing; boloka direkhoto tša motho ka noši e le sephiri.

Ka morago ga 1-5

Hwetša Gae le mohlami. Kgetha sprite ye e nepagetšego. Swantšha ketsakgato ya go thoma le tiro. Akanya go boeletša. Busetša boleng bja mathomo.

Ka morago ga 6-10

Kgetha media ka tumelelo. Hlalosa phapano magareng ga sprite le motheo. Dira diaparo tše pedi tše di swanago. Hlalosa phetogelo ya lefelo.

Ka morago ga 11-14

Swantšha molaetša wa ka gare. Fa tsela ye nngwe ya go tsenya ka sensara. Diriša loop go gogela ka mehla. Leka ka disprite tše pedi.

Ka morago ga 15-18

Busetša le go fetola bolokwa bjo bo fetogago. Hlalosa kgetho e tee. Lemoga gore kgetho ye e sa beakanywago e ka boeletša. Diriša le go kwešiša boloko bjo bo ikgethilego.

Ka morago ga 19-20

Kgetha magareng ga faele ye e ka lokišwago le rekhoto. Tloša dilo tša poraefete. Hlalosa projeke. Hlalosa kaonafatšo e tee yeo e dirilwego ka morago ga ge motho yo mongwe a e lekile.

STEAM e mošomong

Saense: lebelela dintlha tša seswantšho gomme o bapetše ditebelelo le dipoelo. Theknolotši: kwešiša phapano ya media le disedirišwa. Boenjiniere: beakanya, aga, leka le go fetola tirišano. Bokgabo: thala, ngwala, anela le go hlama go babogedi. Dipalo: bala dikhutlo, diphetogo, dibophara le dipheo. Tlhokomelo ya batho, mafelo le bohwa bjo bo phelago e nea diprojeke maikemišetšo a a abelanwago; bokgabo bo rwala tlhaloso ya tšona.

Tlhaloso ya gago ye nnyane ya mantšu

Motheo / Lefelo leo le lego ka morago ga disprite.

Boloko / Taelo e tee goba boleng bo bo tee go app.

Kgetho / Teko ye e nago le karabo ya nnete goba maaka.

Seaparo / Sebopego se sengwe goba maemo a sprite e tee.

Boloko bjo bo ikgethilego / Tatelano ye e filwego leina yeo o e hlalosago go sprite e tee.

Ketsakgato / Selo seo se thomago setaki: play, go thinta, molaetša goba sa go swana.

Romela ka ntle / Dira kopi ka ntle ga app, bjalo ka vidio goba faele ya projeke.

Loop / Khoutu yeo e boeletšago. Boeletša e a bala; Ka mehla e tšwela pele go fihlela e emišwa.

Molaetša / Leswao magareng ga disprite. Beam ke leswao le le fapanego la disedirišwa tše di kgauswi.

Morepoti / Boloko bjo bo kgokologilego bjo bo nago le boleng ka gare ga boloko bjo bongwe.

Busetša / Fa projeke boleng bja yona bja mathomo gape.

Tatelano / Ditaelo tše di beakantšwego ka tatelano.

Sprite / Sebopego, seswantšho goba selo seo o ka se ngwalelago khoutu.

Setaki / skripte / Dibolokwa tše di kgokagantšwego tšeo di šomago mmogo.

Sethale / Lefelo moo kanegelo e bonalago gona.

Bolokwa bjo bo fetogago / Leina leo le bolokago palo yeo projeke e ka e fetolago.

Mantšu a mararo a go a boloka

Beakanya. Leka. Fetola. O ka se hloke gore kgopolo ya gago ya mathomo e be ya mafelelo.

Mothopo le ditokelo (ka Seisemane): letlakala le le latelago le ka fase le ngwadilwe ka Seisemane gore ditsopolo di dule di sa fetolwe. Dikanegelo di hlamilwe tša mathomo; ga se phetolelo.

Sources and credits

OctoStudio was created by the Lifelong Kindergarten group at MIT Media Lab. To learn more, visit octostudio.org. This is an independent Circuit Kids companion book, not an OctoStudio or MIT publication or endorsement.

Stories, teaching text and schematic diagrams in this book were written for Circuit Kids. The Coding Cards section at the back reproduces the official OctoStudio Coding Cards v3.0 (August 2026) unchanged, under CC BY-NC-SA 4.0, with the credit: OctoStudio was created by the Lifelong Kindergarten group at MIT Media Lab. The code recipes describe documented functions in original project contexts. Original book material is licensed under CC BY-NC-SA 4.0 (see page 2). Production-rights clearance remains a release gate.

Current Reference Guide, July 2026, v1.1.0+ https://resources.octostudio.org/ReferenceGuide.pdf

Reference-guide landing page https://octostudio.org/support/learning-resources/octostudio-reference-guide

Learning resources https://octostudio.org/learn/

OctoStudio in Schools https://octostudio.org/learn/octostudio-in-schools

Bring a Book to Life https://octostudio.org/learn/bring-a-book-to-life

Age guidance: especially ages 7-12 https://octostudio.org/support/faq/what-age-group-is-octostudio-designed-for

Offline guidance https://octostudio.org/support/faq/do-i-need-an-internet-connection-and-data-to-use-octostudio

App privacy policy https://octostudio.org/privacy-app

Primary sources: official documentation, official educational resources and published policy/FAQ. No physical-device trial was possible for this edition. Older search results can surface the July 2025 guide, which omits newer costume features.

Where the stories come from

Places and facts in the stories come from public sources: Modjadji Nature Reserve, https://en.wikipedia.org/wiki/Modjadji_Nature_Reserve ; Magoebaskloof, https://en.wikipedia.org/wiki/Magoebaskloof ; Bakone Malapa Open-Air Museum, https://limpopotourism.penit.co.za/destination/bakone-malapa/ ; Polokwane Game Reserve, https://www.lonelyplanet.com/points-of-inter

est/polokwane-game-reserve/1309561 ; Mapungubwe, https://whc.unesco.org/en/list/1099/ . The Modjadji rain queens in Story 17 are mentioned only as something the old people say. Characters, journeys and conversations are fiction. No venue or community endorsement is claimed.

Rights and testing notes

Do not reuse a license by accident

The OctoStudio Coding Cards v3.0 PDF (August 2026) states CC BY-NC-SA 4.0 and allows translation, and this book includes them in that spirit with the required credit. Its landing page has been seen to state Attribution-ShareAlike, so ask the publisher to clarify before using the cards in a paid print bundle.

The official reference guide states Attribution-ShareAlike. This book uses it as technical documentation, not as reproduced page content. If future revisions copy or adapt official diagrams, screenshots or passages, keep an exact asset-and-license ledger and obtain a rights review before adding those assets or selling print copies.

Coding Cards source: https://octostudio.org/support/learning-resources/octostudio-coding-cards and https://resources.octostudio.org/OctoStudio-Coding-Cards.pdf

Learning and print checks

- This edition is for ages 7-12, with a Limpopo family and a wire python that travels with them.
- Project 20 retains an extra testing and reflection section.
- Read and make together. Talk about the stories, care for places, and the difference between new fiction and traditional accounts.
- Test projects on the devices you will use before teaching a group. Check the expected results and adapt the pace to your learners.
- CC BY-NC-SA 4.0 applies to this book. Keep the source-rights record with it.

Circuit Kids | Registration in progress | 072 123 1490 | info@circuitkids.co.za

Dikarata tša Khoutu

Dikarata tša Khoutu tša OctoStudio tše di bontšha ka fao bloko le sehlopha se sengwe le se sengwe di šomago ka gona. Di šomiše go swana le dipheti tša diprojeke.

Dikarata di abelanwa ka tlase ga laesense ya Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0), yeo e dumelelago go fetolelwa. Diswantšho tša dikarata ke tša dikarata tša mathomo; mantšu a ngwadilwego ka tlase ga tšona ke a phetolelo. Maina a dibloko a dula ka Seesemane bjalo ka ge a bonagala ka OctoStudio.

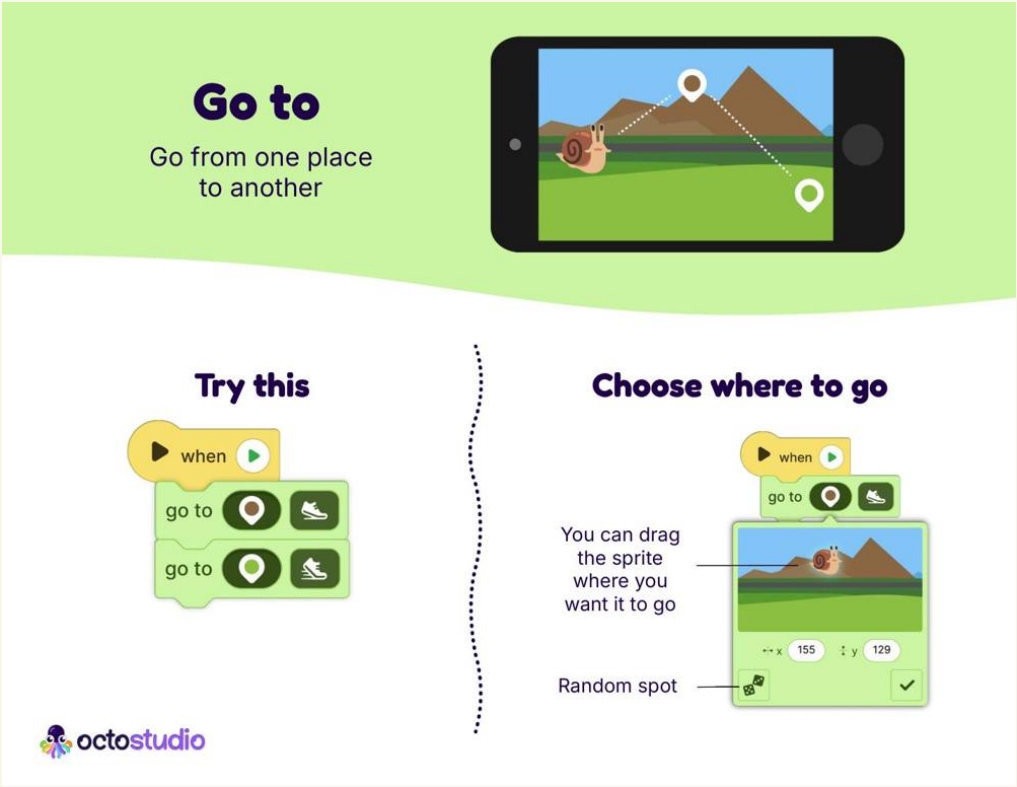
OctoStudio was created by the Lifelong Kindergarten group at MIT Media Lab.

Cards v3.0, CC BY-NC-SA 4.0.

Cards v3.0 (August 2026) - https://octostudio.org/support/learning-resources/octostudio-coding-cards

Eya go

Tloga felong le lengwe o ye go le lengwe

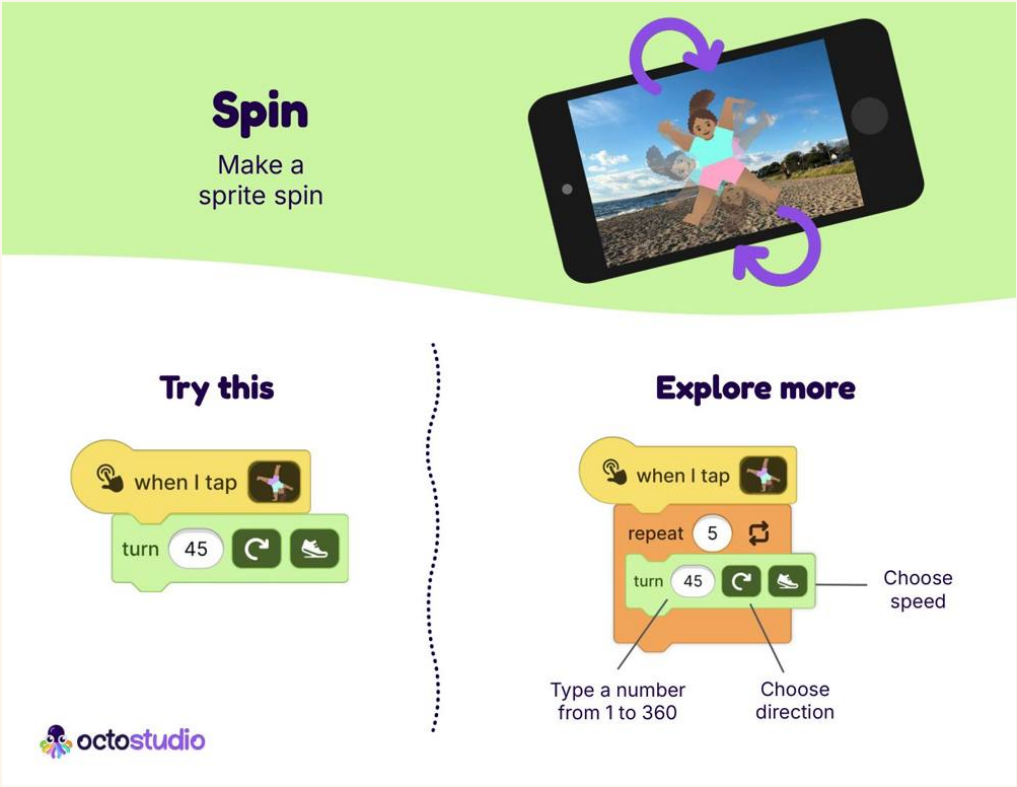


Karata ya mathomo ya OctoStudio (Seesemane).

- Leka se
- Kgetha moo o yago gona
- O ka goga sprite go ya moo o nyakago gore e ye gona
- Felo leo le sa beakanywago

Retolla

Dira gore sprite e retolle

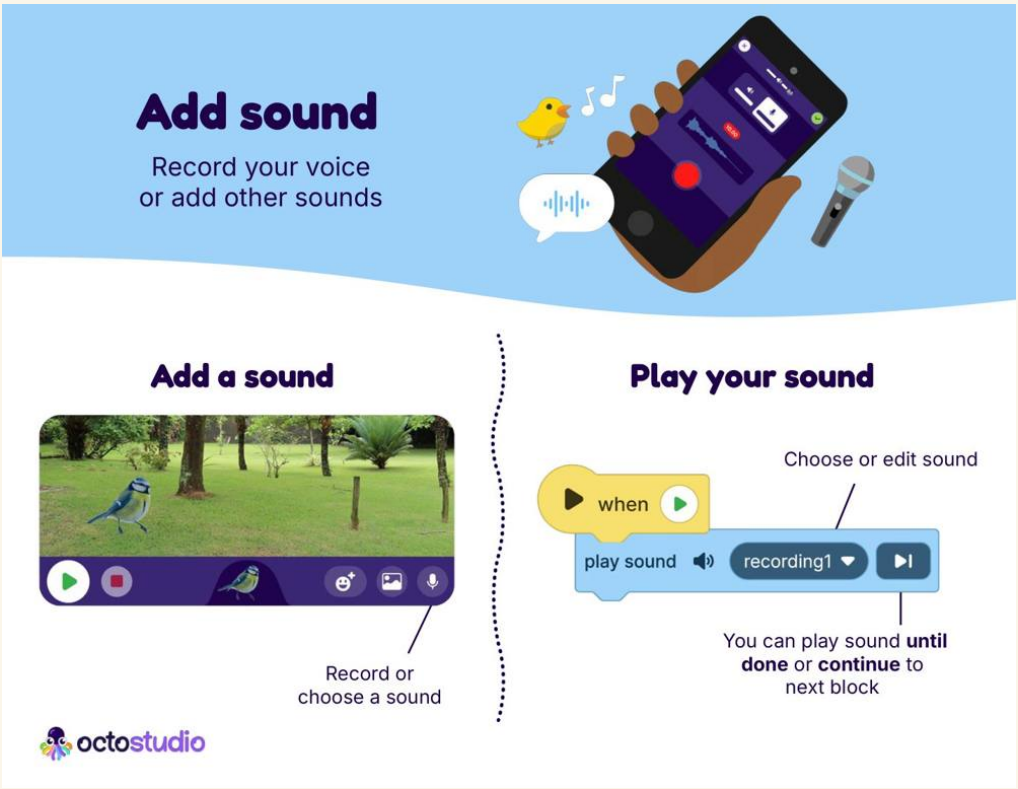


Karata ya mathomo ya OctoStudio (Seesemane).

- Leka se
- Hlahloba go feta
- Kgetha lebelo
- Ngwala palo go tloga go 1 go ya go 360
- Kgetha mothalo

Oketša modumo

Rekhota lentšu la gago goba o oketše medumo ye mengwe

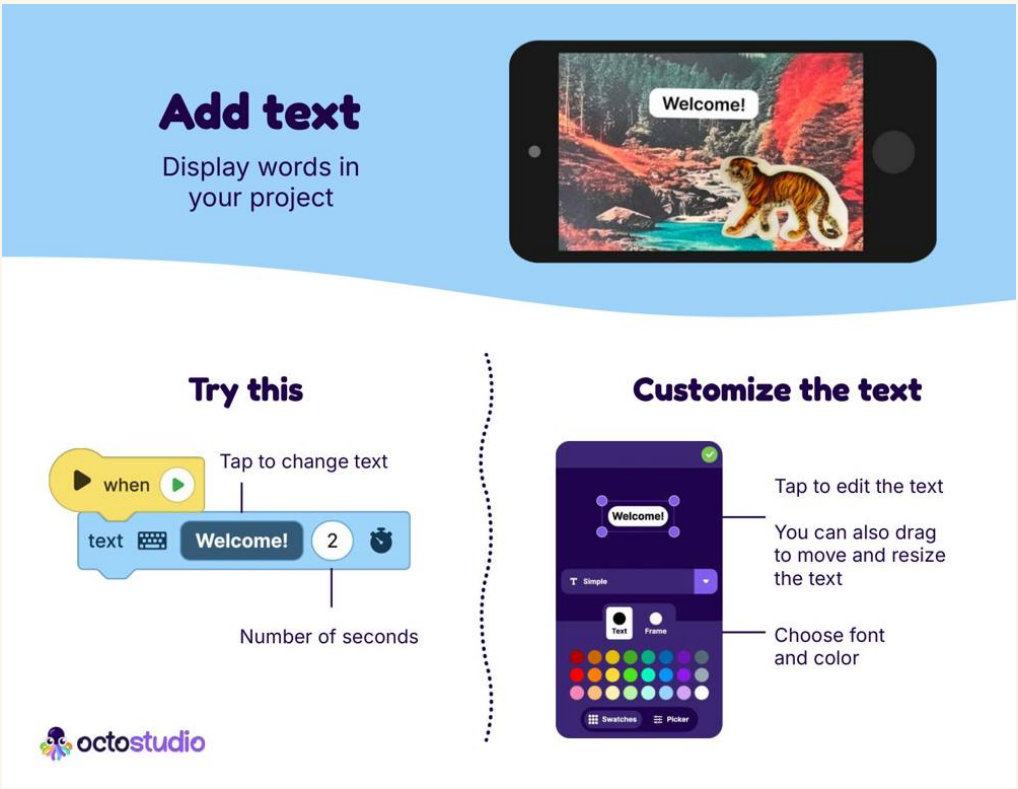


Karata ya mathomo ya OctoStudio (Seesemane).

- Oketša modumo
- Letša modumo wa gago
- Kgetha goba o lokiše modumo
- Rekhota goba o kgethe modumo
- O ka letša modumo go fihlela o fela goba wa tšwela pele go boloko bjo bo latelago

Oketša mongwalo

Bontšha mantšu projekeng ya gago

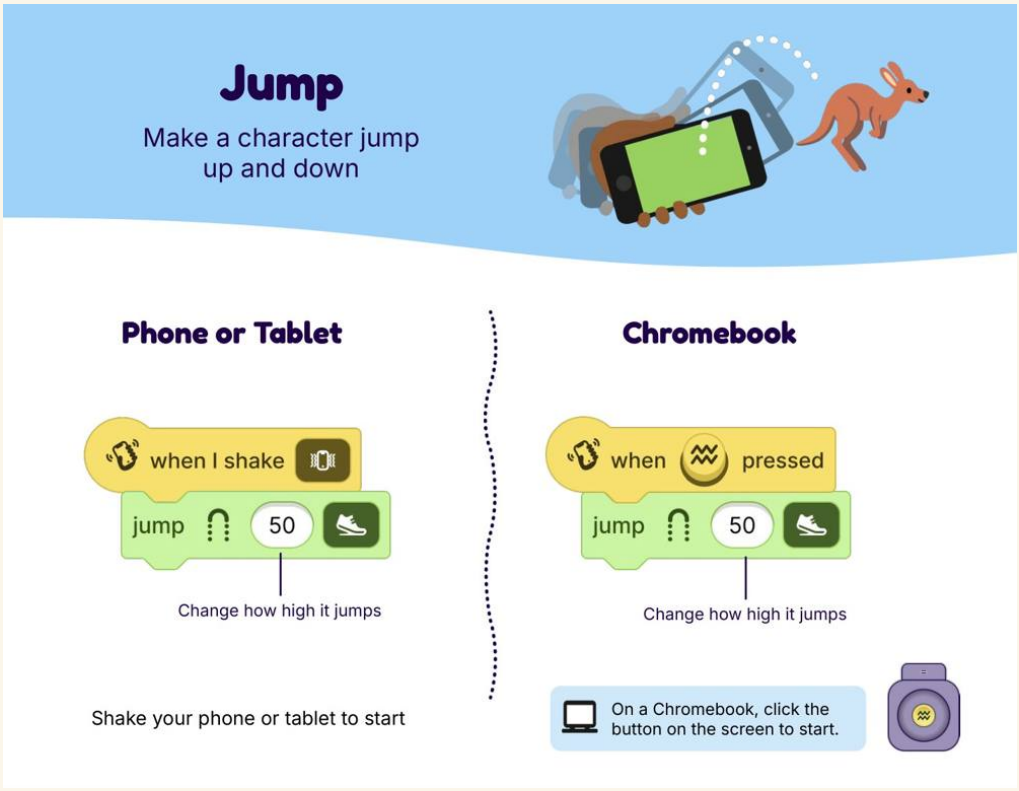


Karata ya mathomo ya OctoStudio (Seesemane).

- Leka se
- Lokiša mongwalo
- Thinta go fetola mongwalo
- Thinta go lokiša mongwalo
- O ka goga gape go o šuthiša le go fetola bogolo bja gona
- Palo ya metsotswana
- Kgetha mohuta wa mongwalo le mmala

Tlola

Dira gore sebopego se tlole godimo le fase



Karata ya mathomo ya OctoStudio (Seesemane).

- Mogala goba Tablet
- Fetola gore e tlola godimo go kae
- Šikinya mogala goba tablet ya gago go thoma
- Chromebook
- Go Chromebook, thinta konope ye e lego sekirining go thoma.

Gola le go fokotšega

Dira gore sprite e gole goba e fokotšege

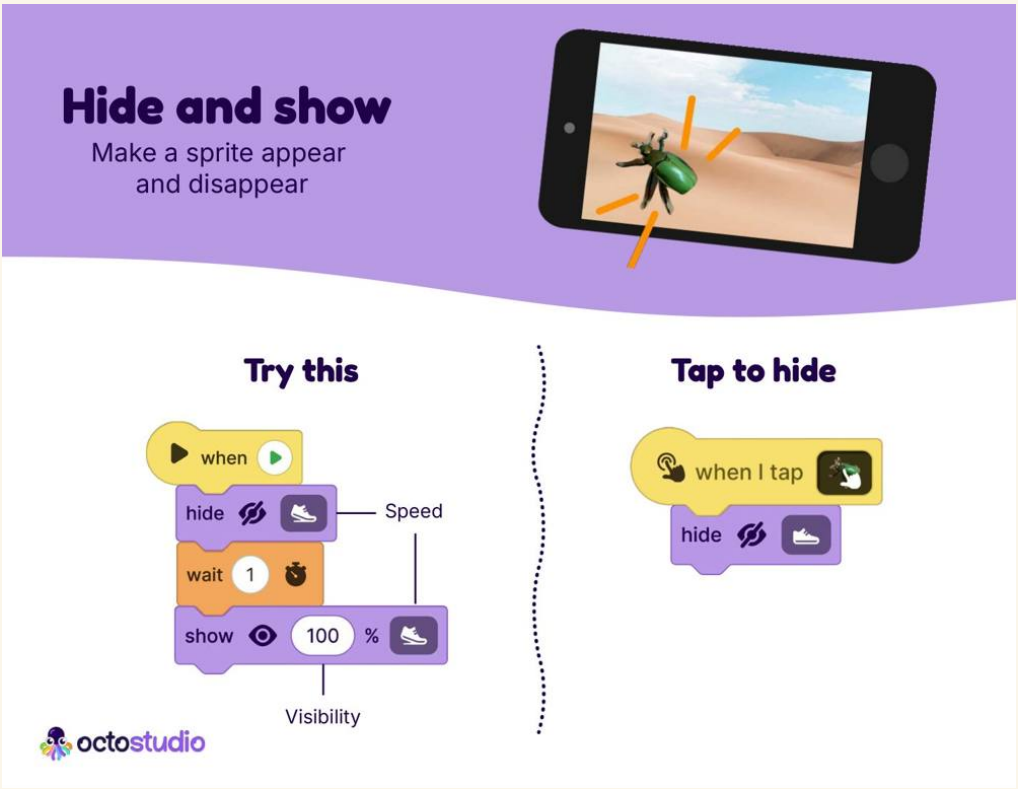


Karata ya mathomo ya OctoStudio (Seesemane).

- Leka se
- Leka dipalo tše di fapanego
- Oketša go feta

Uta le go bontšha

Dira gore sprite e bonale gomme e nyamele

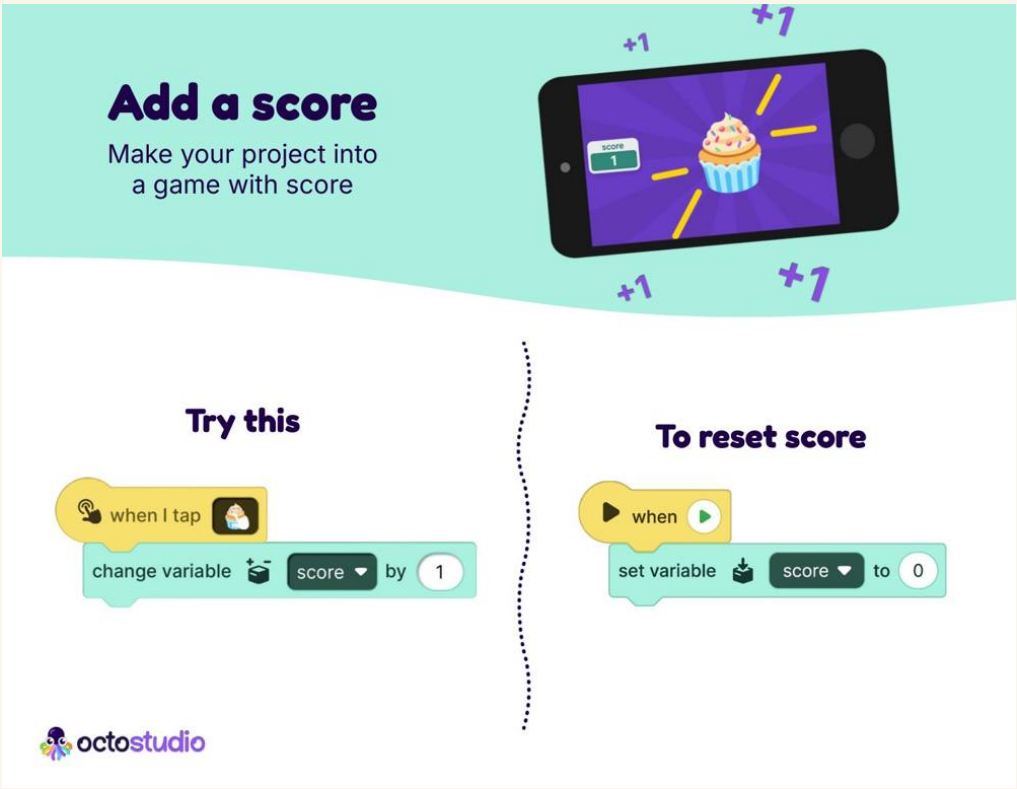


Karata ya mathomo ya OctoStudio (Seesemane).

- Leka se
- Thinta go uta
- Lebelo
- Go bonagala

Oketša dintlha

Dira projeke ya gago papadi ye e nago le dintlha

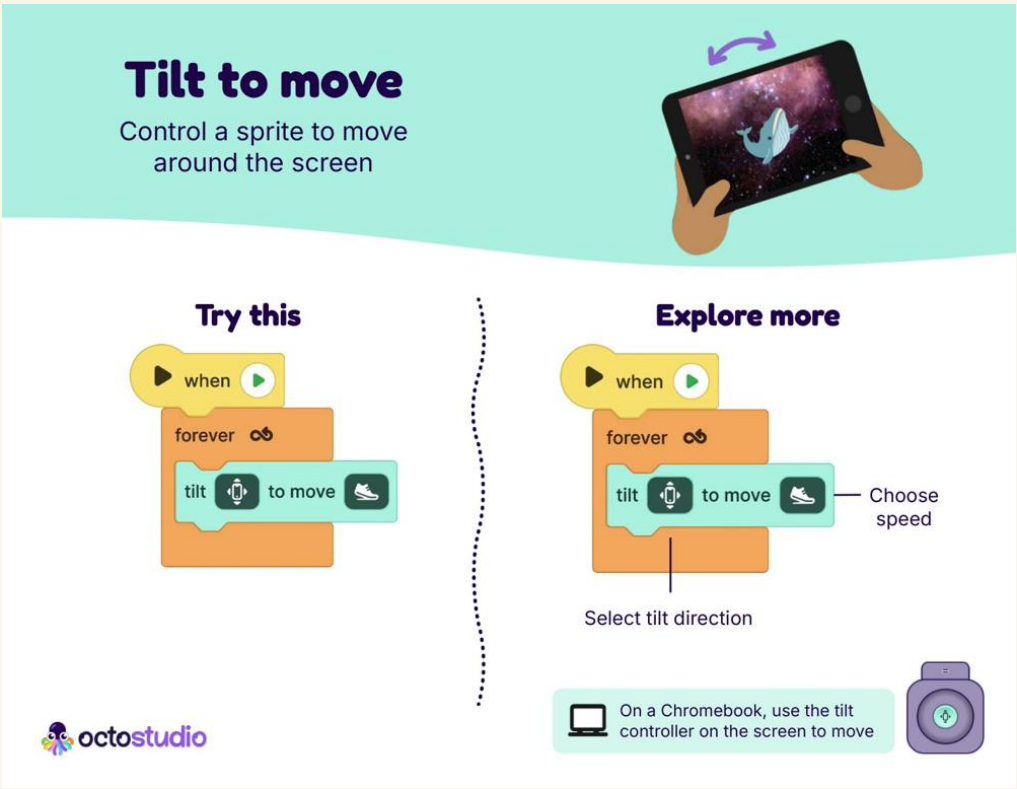


Karata ya mathomo ya OctoStudio (Seesemane).

- Leka se
- Go busetša dintlha

Sega go sepela

Laola sprite gore e sepele sekirining

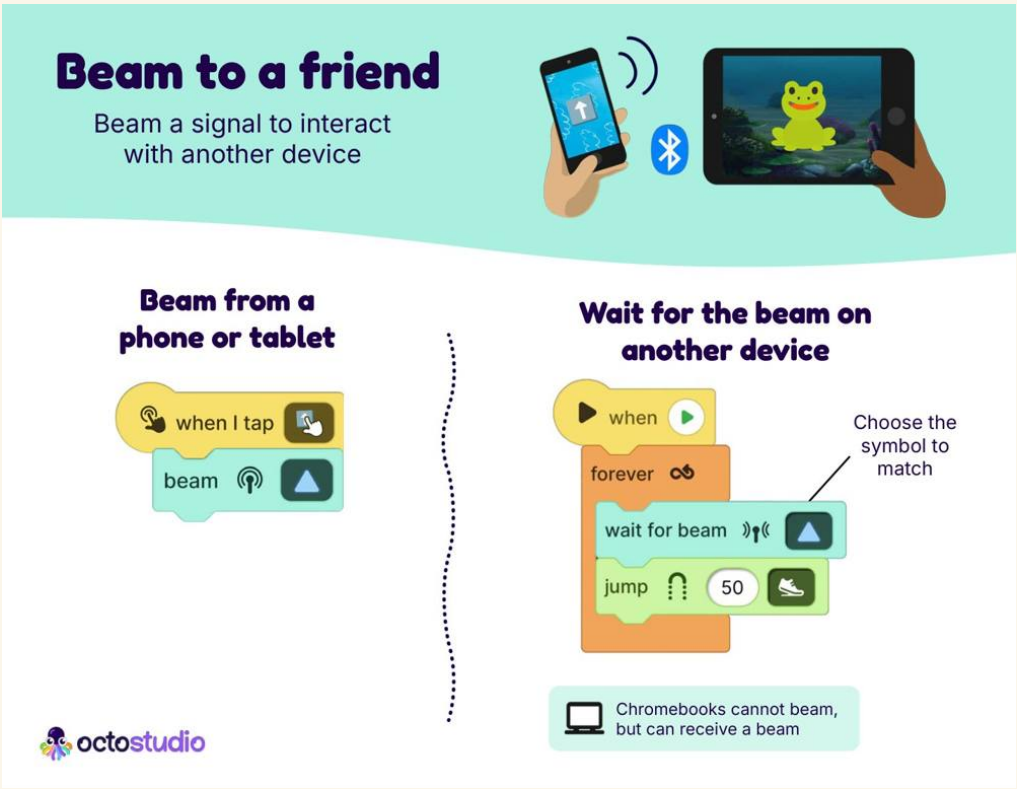


Karata ya mathomo ya OctoStudio (Seesemane).

- Leka se
- Hlahloba go feta
- Kgetha lebelo
- Kgetha mothalo wa go sega
- Go Chromebook, diriša taolo ya go sega ya sekirini go sepela

Beam go tsala

Romela leswao go dirišana le sedirišwa se sengwe



Karata ya mathomo ya OctoStudio (Seesemane).

- Beam go tšwa mogaleng goba tabletheng
- Leta beam sedirišweng se sengwe
- Kgetha leswao leo le swanago
- Chromebooks ga di kgone go beam, eupša di ka amogela beam

Bea seaparo

Fetoga go tšwa seapareng se sengwe go ya go se sengwe

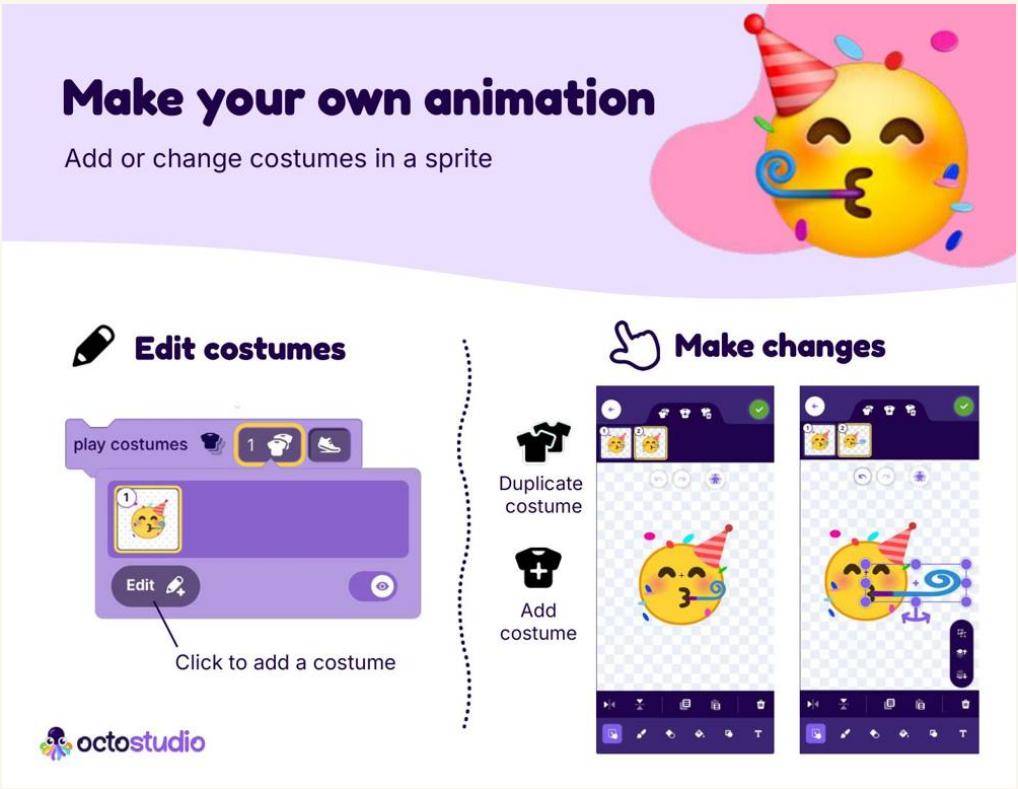


Karata ya mathomo ya OctoStudio (Seesemane).

- Bula lenaneo go kgetha seaparo
- Fetoga go tšwa seapareng se sengwe go ya go se sengwe
- Kgetha lebelo

Dira animeišene ya gago

Oketša goba o fetole diaparo sprite

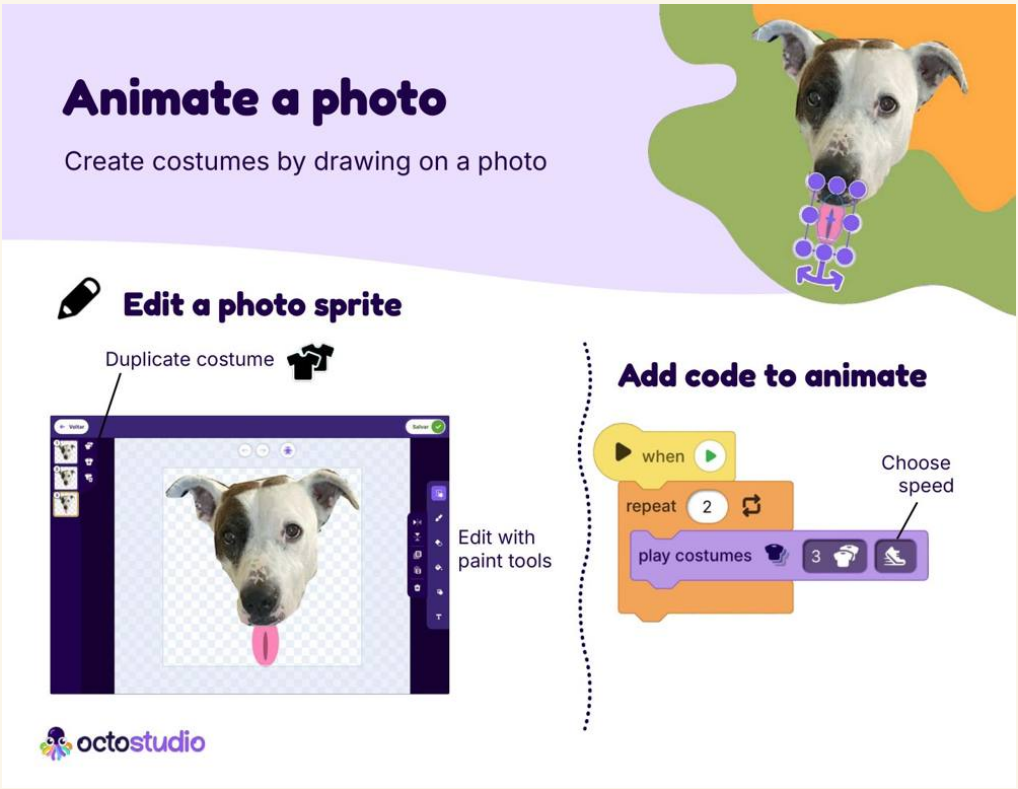


Karata ya mathomo ya OctoStudio (Seesemane).

- Dira diphetogo
- Lokiša diaparo
- Kopa seaparo
- Oketša seaparo
- Thinta go oketša seaparo

Dira seswantšho se sepele

Hlama diaparo ka go thala seswantšhong



Karata ya mathomo ya OctoStudio (Seesemane).

- Lokiša sprite ya seswantšho
- Kopa seaparo
- Oketša khoutu go dira motšamao
- Kgetha lebelo
- Lokiša ka didirišwa tša pente

KARATA 14 / 14

Motšamao

Dira gore sprite e sepele, e fofe goba e rotoge



Karata ya mathomo ya OctoStudio (Seesemane).

- Kgetha sprite ye e nago le diaparo
- Dira motšamao ge e sa sepela
- Diphoofolo le batho ba bantši ba sprite ba tla le seaparo se tee goba go feta.
- O ka diriša ditaki tše pedi goba go feta tša dibolokwa go thoma ka nako e tee.
- Kgetha lebelo